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The Player Experience and Design Implications of Narrative Games

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ABSTRACT

Narrative games focus on the plot, character interactions, and sometimes have story variations depending on how the player interacts with them. This study adopts user experience approaches, such as interviews, think-aloud protocol, and grounded theory, to examine the player experience of commercial narrative games. We identify four types of players' interactional patterns that include exploring the story, projecting the story, playing the game, and watching the story. Based on the study of the player experience, we derive a set of principles from these interactional patterns that game designers can draw upon when designing narrative games. These principles include player-avatar dynamic, a strong interactive space for the player to explore, placeholder for projecting an emergent narrative, inclusive game design to enhance accessibility, using an interactive feedback loop to adapt to the players' playstyle, quick-time events and emotional connection to the characters, and more complex options to romance/relationships in narrative games.

1. Interactive digital narrative

Interactive stories in the form of narrative games can be powerful agents of personal transformation by offering users the opportunity to enact stories rather than just watch them unfold onscreen (Murray, 2017). In *Hamlet on the Holodeck*, Janet Murray introduces the notion of the “multiform” story to convey the idea of multiple parallel possibilities that can be created from our choices, and that can usually reflect different perspectives of the same event, when using an interactive work such as a digital life story (Chew & Mitchell, 2019). The term “multiform story” refers to stories that exist as variations of a single scenario, and in a well-constructed multiform story, each individual instantiation stands alone as a coherent narrative (Murray, 2017). A multiform story is designed to motivate the user to replay the same scenario to access the deeper implications that can emerge from well-formed variations (Murray, 2017). The multiform story is also related to other similar work on virtual and parallel storyworlds such as interactive stories (e.g., Mitchell & McGee, 2011), interactive fiction games (e.g., Sharma et al., 2010), hypertext fiction (e.g., Shin, 2017), experimental film narratives (e.g., Hales, 2020), documentary film (e.g., Alexis, 2021), visual novels (e.g., Wang et al., 2021), and video games (e.g., Ostenson, 2013). An example of a multiform story in an electronic game is *The Witcher 2 (TW2)*, in which two major paths are offered to the player over the course of the game—the human path or the elven path—depending on the player's choice of ally.

The study of the player experience of the interactive story in narrative games is important because it can help to

illuminate the specific types of interactional patterns in their engagement with the story. In Toh's (2018) study, the player experience is defined as the participants' understanding of the game structure as a whole, which consists of the different relationships between narrative and gameplay in the ludonarrative model and contextual factors such as the player's prior gaming experience. The player's narrative consists of the participants' cognitive interpretations and the gameplay column in the ludonarrative model comprises the participants' knowledge of the gameplay rules, which allows them to understand what they are allowed to do in the game. The player experience of the different ludonarrative relationships allows the participants to experience different emotions when playing the game. For instance, in ludonarrative resonance, which is defined as the extreme fit between gameplay and narrative (Toh, 2018), the player can experience the emotions of satisfaction and excitement when they immerse themselves into the flow of the game and have an extreme interaction experience (VandenBerghe, 2012) with well-designed narrative games such as *The Last of Us (TLOU)* and *The Walking Dead (TWD)* containing a story that they want to believe in. They can experience empathy to understand the thoughts, emotions, and behaviour of the player character (PC) (Toh & Lim, 2021) they are role-playing as when the actions that they can perform using the PC align with how the PC is being represented in the game's narrative such that they can feel that they are playing an active part in the game as the story unfolds.

A deeper analysis of the types of interactional patterns, which is carried out in this paper, can help us to derive

design principles to allow game designers to better create interactive works that improve the user experience. The research on interactive digital narratives (IDN) is ongoing (see Koenitz, 2015; Mitchell et al., 2020; Mitchell & Kway, 2020; Roth & Koenitz, 2017; Sim & Mitchell, 2017; Tan & Mitchell, 2020). Although related research on the player experience and narrative forms in games has been conducted including Brenden Keogh's (2018) work on bodies and the player experience, Darshana Jayemanne's (2017) work on performativity in games, and Cody Mejeur's (2020) work on visualizing different player experiences in games, much work has yet to be done. In this paper, we build on the work of IDN by analyzing the player experience of narrative games in the form of their interactional patterns to derive a set of design principles for game designers to consider when making games. The analysis of the player experience in this paper expands research on IDN and addresses gaps in the ludology versus narratology debate (c.f., Mukherjee, 2015), research conducted on narrative in games, patterns in games, and human computer interaction research conducted on intermediate knowledge representations by providing empirical data of different players to generate the types of interactional patterns they exhibit during gameplay. Before I review the relevant literature related to this study, I will provide some background on how narrative video game design has evolved over the years and how technology has created a positive impact on storytelling and game mechanics to contextualize the study.

1.1. Evolution of narrative game design

The earliest video games in the 1970s such as *Spacewar*, *Computer Space*, and *Pong* had very rudimentary graphics and the games' narrative was very minimal or non-existent due to the limitations of technology during that time. The players usually have a single objective in early video games, which is to gain as many points as possible by controlling the game sprite to move across the screen to hit an objective. In the 1980s, there were some improvements in the technology used to create and play games. For instance, the development of video game display technology such as liquid-crystal display (LCD) for handheld devices enabled visuals to be displayed on portable game sets and was commonly found in Nintendo's "Game and Watch" series. However, the LCD displays were limited in their functionality as they hold a pre-formed set of visuals to represent the visual data in a set game. This type of display was also limited to smaller screens and simple games. Vector and raster graphics were also commonly used in 1980s video games. Vector graphics used the x, y coordinate system with games and raster graphics provided pixels with a limited spectrum of colours to display the visuals. These technologies allowed the visuals in narrative games during the 1980s to be displayed as two-dimensional pixelated objects and in a restricted range of colours.

In the 1980s, games such as *Donkey Kong* and *Alley Cat* consisted of visuals with two-dimensional models and a limited range of colours was used to display the game world

and characters found in it. There was also a narrative created to accompany the gameplay. However, the narrative was very linear and served as a context for the gameplay. Cutscreens were used at the beginning of the game to contextualize the game's story and at the end of each of the game levels to communicate the narrative progress and motivate the player to continue playing the game. The plot was communicated using small animations and text such as "help" in *Donkey Kong* to represent the damsel calling for assistance and "alas!" in *Alley Cat* when the PC hit the spider found inside the people's home and had to start over outside the house. The game mechanics which included points awarded to the players as they completed each level were very simple and the gameplay comprised of sequential levels that the players were to complete in a fixed linear order.

Sierra Entertainment was the first few companies to pioneer narrative games in the 1980s. It is known for pioneering the text-based graphic adventure game genre, which includes the text-based graphic adventure game series *King's Quest*, *Space Quest*, *Police Quest*, *Leisure Suit Larry*, and *Quest for Glory*. Although these games do not truly adopt a full multi-linear narrative and multiform gameplay, they allow players to have a sense of multiform and non-linear progression. For instance, in *Quest for Glory*, each character class has its own unique way to solve different in-game puzzles and some puzzles have up to four different solutions. Each class also has specific quests and storylines in each game that focus on their respective guilds, and thus the games are designed for replay value (Kaiser, 2012). *King's Quest* also has multiform gameplay, and the player can choose to collect the three legendary treasures in the storyline in any order, even though the player will ultimately end up in the same place (Salter, 2019). *Police Quest* follows a linear storyline, but there is multiform gameplay in which the player is free to explore Lytton's bars, parks, hotels, and courtrooms at their own pace (Nguyen, 2015). Like *Police Quest*, *Leisure Suit Larry* follows a linear storyline with minor variations in each playthrough. The player can engage in the multiform gameplay by exploring the bar, casino, wedding chapel, disco, and a convenience store at their own pace to find gifts and win money to woo the women in the game and advance to the game's final area, the hotel penthouse. Finally, *Space Quest* has also a largely linear narrative with minor variations and the multiform gameplay is present such that it allows more than one way for the player (to die or) to solve the puzzles and overcome the challenges in the game.

Despite the limitations of technology in the 1980s, one influential game that is worth mentioning is the role-playing game, *Ultima*, first released in June 1981. It is influential in revolutionising narrative game design by introducing open-world gaming in which the PC can travel across swamps, oceans, and hills to discover what the game world had to offer. In other words, player exploration can proceed in a multi-linear manner leading to varied narrative possibilities. *Ultima* was also at the forefront of plotting in video games. It was one of the first games to have the player control one

specific character, the Avatar, which the player created along with other characters that already existed with pre-determined roles and personalities, who can be engaged for conversation. *Ultima IV* is another early game that shows us how the game's story can be told through its gameplay. For instance, the actions that the player performs in the game will affect the virtue scores of the PC, the Avatar in the story. These cases show that technology is not the limiting factor in creating innovation in narrative game design.

The 1990s was a decade of marked improvements in video gaming technology in which there was a transition from sprite-based two-dimensional graphics to full-fledged three-dimensional graphics. The improvements in video game technology gave rise to several genres of video games, such as racing games, first-person shooters, real-time strategy games, survival horror games, puzzle games, and so on. However, the story in these video games was still quite linear, although it depends on the genre. For instance, the first-person shooter *Wolfenstein 3D* has a linear scripted story and a linear gameplay with no branching options. The increase in storage space and computing power enabled narrative game design to include more elaborated plots and to incorporate multiple endings. Some game genres such as the adventure puzzle game, *Myst*, the fighting game, *Street Fighter Alpha 3*, the role-playing game, *Fallout 2*, and the action role-playing game, *Star Ocean: The Second Story* have multiple narrative endings depending on the player's choices and actions.

The 2000s saw the last of the low-resolution three-dimensional polygons of the 1990s with the emergence of high-definition games. The further advancement of technology facilitated the development of immersive and interactive environments, the implementation of realistic physics, and the improvement of artificial intelligence in games. The development of hardware with even more storage space and increased memory capacity of gaming systems also contributed to the development of narrative games with an even more expansive plot comprising of the main storyline and several side plots of multiple characters. Technological improvements contributed to the expansion of game narratives with multiple endings to more game genres such as the first-person shooter, *Bioshock*. The players can immerse in the game world by moving across the interactive game space and interacting with different characters and game objects in the games released in this period to experience different types of emotions (e.g., Cole, 2015; Ebadi & Ahmadi, 2022). However, there were limitations in the technology used to deliver story content to the players, such as the computer-generated facial expressions of the characters and reusable character models in *Mass Effect (ME)* and *Bioshock*,¹ which led to participants 6 and 9's perception of contrived character facial expressions and movements when they played the games in the study. Some other games released in the 2000s included *Shadow of the Colossus*, *The Elder Scrolls IV: Oblivion*, *Silent Hill 2*, and *Dwarf Fortress*. *Dwarf Fortress* is an indie game released in 2006 and uses procedural generation techniques to create gameplay content. The game adopts text-based graphics and is open-ended with no set

objectives. It does not have a scripted narrative and it is the emergent narrative produced by *Dwarf Fortress* players across different media that adds to the player experience of this video game (Adams, 2018, 2021).

Finally, the period from 2010s to 2020s in narrative game design saw even further innovation fuelled by the technological advancements, such as virtual reality, augmented reality, and mobile devices including tablets and mobile handheld phones that serve to enhance the player experience. For instance, the use of virtual reality technology in narrative game design has been found to enhance participants' game immersion and empathy towards the non-player characters (NPCs) in the game (Ho & Ng, 2022) and augmented reality can result in immersion, emotional involvement, and enjoyment in a mobile game (Sekhavat & Zarei, 2018). The development and implementation of performance and motion capture technology in narrative game design enables video game characters and their facial expressions, gestures, and character movements to be modelled on real-life actors, such as those found in *The Last of Us*, *Beyond: Two Souls (BTS)* (Cook, 2018), *LA Noire* (Pallant, 2013), and *The Witcher 2* (motion capture² used in the cutscenes³ and animations for dialogue, etc.). Performance and motion capture technology helps video games to deliver emotional content to the players more effectively through the characters' facial expressions and more realistic movements to allow players to connect emotionally with the characters and immerse in the game's story. The implementation of procedural storytelling models (e.g., Short & Adams, 2019) into the game design also facilitated the creation of video games with game narratives that can change according to player choices. Games that feature narrative that can adapt to the player choices include *The Walking Dead*, *The Witcher* videogame series, *Beyond: Two Souls*, and *Life is Strange*. The game narrative's ability to adapt to the player choices engages and immerses the player in the story world of the game when the player feels that the choices that they make during gameplay are meaningful (Iten et al., 2018) and can affect the game's narrative in an intentional manner (Tanenbaum & Tanenbaum, 2010).

1.2. Ludology versus narratology debate

The ludology versus narratology debate is an important milestone in game studies (Frasca, 2003; Mäyrä, 2008) in which gameplay and narrative are conceptualized as either competing (e.g., Eskelinen, 2001) or complementary perspectives (Kapell, 2016; McManus & Feinstein, 2006). Early on in the debate, game studies was presumed to be a divided turf, with narratologists who believed that games are used for telling stories, on one side, and ludologists who advocated for the study of play and its mechanics, on the other side (Frasca, 2003). This dichotomy was however an oversimplification. During the height of the ludology versus narratology debate in game studies, Frasca (2003) identified "narrativist" as a third position between literature-derived narratology and rule-focused ludology (Toh, 2015). The narrativist approach focuses on creating digital-native narratives

where interactivity is the core property and is not an addition to long established literature-based narrative structures (Toh, 2015). The importance of the debate is in demonstrating the value of combining narrative and gameplay in narrative game design to engage the players, immerse them in the game, and allow them to form emotional connections to the game characters in their player experience. Over the years many game scholars have shown the benefits that different disciplines and approaches bring to understanding games (Salter, 2019). Game designers have also seen the value in a playable narrative, which has been found in narrative games such as adventure games. Game designers have continued to take up the challenge of building a new future for narrative games by rethinking the possibilities for fusing story and interactivity (Salter, 2019).

Moving on from the debate, arguments have been made for the integration of narratological and ludological perspectives, for example, in the study of ideology (Pérez-Latorre & Oliva, 2019), interactive narrative game design (e.g., Larsen & Schoenau-Fog, 2016, 2019; Ralph & Monu, 2015; Sim & Mitchell, 2017), the player experience (Roth et al., 2018; Toh, 2018), political critique (Murphy, 2016), empathy play (Kuznetsova, 2017), and as a pedagogical tool (Walther, 2019). Going beyond the narratology and ludology perspectives, narrative theory has been extended to cover emergent narrative in AI-based experiences and video games (e.g., Louchart et al., 2008; Louchart & Aylett, 2004) and the multimodal approach to video game analysis focuses on our meaning-making of the elements (and modes) within the text as a whole (e.g., Toh, 2018; Hawreliak, 2018; Hawreliak & Lemieux, 2020; Ng, 2019, 2020; Stamenković et al., 2017).

Rather than advancing the notion that the gameplay and the narrative are discrete categorical concepts (e.g., Pinchbeck, 2009) and hypothetical constructs (e.g., Aarseth, 2012; Eskelinen, 2012) based on the ludology versus narratology debate, we follow the principle that human-computer interaction is structured in a series of complex dynamic interactional patterns (Kovanen et al., 2013) that change along with the players' mental models (Toh, 2018) as they learn about the game during the process of gameplay.

1.3. Narrative in games

There is a large body of work on narrative in games (for a review, see Toh, 2018). Related to our study is the examination of poetic gameplay devices in interactive life stories (Chew & Mitchell, 2020), the comparative content analysis of a corpus of autobiographical games (Werning, 2017), and the exploration of narrative design and the player's potential aesthetic response in games like *The Last of Us* (Farca & Ladevèze, 2016), *Bioshock Infinite* (Farca, 2018) and *The Walking Dead* (Smethurst & Craps, 2015). Other related works include Ryan's (2009) typology of interactive narrative, Katsarov et al.'s (2019) evaluation of 20 distinct game mechanics of narrative games such as *Bioshock* in view of their potential to promote players' moral sensitivity, and the discussion of the relationships between the ludic and narrative elements in narrative-heavy games like *Beyond: Two*

Souls (Riddle, 2016). However, there is no empirical data of different players to support the analyses made in these studies.

Flamma (2016) critiqued the design features of narrative games like *Beyond: Two Souls* from both players' and researchers' perspectives, but did not conduct an in-depth systematic analysis of the games by using member checking (Birt et al., 2016) to triangulate the researchers' experiences with different players' experiences to explore the credibility of results. Taylor et al.'s (2015) microethnographic exploration of eight participants' dynamic engagements in *The Walking Dead* was descriptive and subjective, and the findings may not be representative. In their experimental study on a single role-playing game, *Mass Effect 2* Moser and Fang (2015) found that branching narrative structure led to the participants' improved enjoyment and flow experience and the number of salient decisions had augmented the participants' enjoyment related to characters and cognition in the branching narrative condition. However, the participants were only allowed to view prerecorded gameplay sessions for the purpose of experimental control and their reported experiences may be different if they were allowed to play the game during the experiment. Burgess and Jones' (2020) thematic analysis of *Mass Effect* players' emotional attachments to NPCs in forum posts was not triangulated with interviews and member checks. In our study, we conduct an empirical study of the player experience in six commercial narrative games to come to an overall understanding of the participants' player experience.

1.4. Patterns in games

Game design patterns, including narrative patterns in story-based games (Zubek, 2020), are specific reoccurring parts of the design such as interactions, dynamics (Zubek, 2020), and characteristics that emerge from component configurations of game elements (Björk & Holopainen, 2006), and can be used for the analysis and refinement of narrative game mechanics during the game making process and player experience studies (Toh, 2018). These design patterns are developed by combining insights from different disciplines such as game design with affective learning theories (Dormann et al., 2013) and by combining cognitive narratology with game design theory (Dubbelman, 2017; Thoss et al., 2018; Wilks et al., 2022). Game design patterns are important because they can provide a design template to help in the conceptualization (Dormann et al., 2013) and classification (Schmitz et al., 2013) of the game, help to break up larger game design problems into smaller pieces for troubleshooting (Björk & Holopainen, 2006), or offer a bottom-up approach of designing games (Zavcer et al., 2014).

Using grounded theory, we coded the data from interviews and think-aloud verbalizations in the video recordings (see Section 2.3 data coding) and derived the types of dynamic interactional patterns that players engage in during gameplay. The analysis is situated within the participants' interactional context, including their sociocultural knowledge and the negotiated interaction between the researcher

and participants in the laboratory (Girvan & Savage, 2010) that goes beyond the constraints of conventional game theories (Ang et al., 2010) about narrative.

1.5. Intermediate knowledge representations

Also relevant to our study is the human computer interaction research conducted on intermediate knowledge representations, which can refer to the process of sensemaking and the changes of user mental models of the task or domain over time and the types of changes that occur (Zhang & Soergel, 2016). The notion of patterns, which include the game design patterns discussed in Section 1.4, is an example of intermediate-level abstraction and clearly represents attempts to elicit the key ideas of particular artifacts or families of artifacts (Löwgren, 2013). Experiential qualities also consist of intermediate-level knowledge representations, similar to patterns, but pertaining to users' experiences rather than elements of the artifacts themselves (Löwgren, 2013). As this study examines the player experience, the notion of experiential qualities as intermediate knowledge representations is relevant for our analytical purposes and is conceptualized as the participants' interactional patterns.

Related prior studies on intermediate knowledge representations have focused on the sensemaking process (e.g., Zhang & Soergel, 2016), website resources (e.g., Gray & Kou, 2017), neural networks (e.g., Chen et al., 2019), description logic programs (e.g., Zhu et al., 2010), annotated portfolios (Gaver & Bowers, 2012), and interaction design (e.g., Höök & Löwgren, 2012). The limitation of these studies is that the process patterns and types of conceptual changes uncovered may not be inclusive for all sensemaking tasks as they are conducted in specific domains. In contrast, our study aims to induce the types of interactional patterns of the participants on the level of intermediate knowledge representations based on the analysis of their think-aloud protocol, reflections, interviews, and gameplay recordings to propose design principles for narrative games.

1.6. Our study

Despite the importance of intermediate knowledge representations such as the participants' interactional patterns during gameplay, there is a dearth of research conducted on the different types of interactional patterns in narrative games and how these patterns inform the player experience. Instead of only focusing on the individual actions of the player, we aim to explore the types of interactional patterns of the participants (Toh, 2018; c.f., Mejeur, 2020) in their gameplay recordings and examine how these patterns shift during the process of gameplay. With this aim in mind, we coded the data using grounded theory and the thematic analysis (see Method Section 2.3 for the process) and developed a coding scheme for the games in the study that can be used by game designers to conduct user experience studies. Additionally, the human computer interaction dimensions such as the interactional patterns are not only applicable to video games, but can also be extended to the analysis of

serious games (e.g., Wong et al., 2021), educational games (e.g., González-Ferreras et al., 2017; Yu et al., 2021), training simulations (e.g., Milcent et al., 2022), and so on in future studies. The human computer interaction dimensions in the form of the interactional patterns are important because they influence the proposed recommendations for narrative game designers to a large extent. Building on prior studies of IDN, we examine the players' experience by asking the following research questions:

1. What interactional patterns do players adopt in the games they played?
2. What are the design principles that game designers can learn from the players' interactional patterns in the games they played?

2. Method

2.1. Participant recruitment

The data in this paper is drawn from a study on player experience (Toh, 2018). Participants were recruited through convenience and snowball sampling through advertisements on the university's website and their ages ranged from 19 to 33 years. A total of 37 participants were recruited, but data in this paper were only drawn from the 11 who completed the entire study, which included completing play of the entire video game and participating in interviews as well as one additional player (participant 12) who completed *TW2* but did not complete the interviews. The participants were selected as they possessed at least five years of video game-playing experience and were, therefore, considered expert players (Soylu & Bruning, 2016) who did not require much training to play the selected game(s) for this study. Participants 3, 8, 10, and 12 have played their selected games before, while participants 6 and 9 have watched play-throughs of *TWD* and *TLOU* respectively on YouTube. The rest of the participants were playing the game for the first time. The genres of games these participants were most familiar with were shooters, puzzle games, action games, role-playing games, and strategy games.

2.2. Gameplay sessions

During the first session, the participants completed a written survey to provide information regarding their demographics and gaming habits. Then, we observed the participants playing in the laboratory on a monitor on which the gameplay was being streamed and recorded. They played the game from the beginning for a duration of 1–3 hr. Video recordings of the participants' gameplay were made using Fraps (if they were playing on the computer) and a PlayStation 3 (PS3) recorder (if they were playing on the console), which provided us with a reference when asking the participants to reflect on and verbalize (using the retrospective think-aloud protocol) what they learnt from playing. The participants were given the choice to play on a computer versus PS3 by comfort except for *Beyond: Two Souls* because we did not

manage to recruit enough participants to play this game. After the participants completed their first gameplay session, we conducted and recorded an interview with them, which lasted for approximately 1–2 hr, to understand player experience. We then debriefed the participants by informing them about the aims and nature of the study. The debriefing protocol for the study involved thanking the participants for taking the time to participate in the study, explaining to them in layman terms what was being studied, providing participants with an opportunity to withdraw their consent to participate or to withdraw their data from the study, and informing them that their identity will be anonymized when their data is used for publication.

In subsequent sessions, the PS3 participants who did not have a PS3 at home continued to play in the laboratory by incorporating their natural reactions with their think-aloud verbalizations (using the concurrent think-aloud protocol). The group of computer participants continued to play at home and were taught how to record their gameplay with their think-aloud verbalizations using Fraps. The computer participants were allowed to continue gaming at home because they preferred to do so by comfort. The number of gaming sessions for the participants are listed in Table 1. The number of gaming sessions refers to the number of separate sessions that the participants took to complete the game. After they completed the game (in approximately 7.2–34.6 hr), we reviewed their game recordings and developed specific interview questions to conduct member checks (Cho & Lee, 2014) with the participants by verifying our interpretations of their recordings in the last interview.

Then, we arranged the final interview session to review their gameplay recordings and discuss their game experience by using the interview questions, stimulated recall, and the retrospective think-aloud protocol. In the final interview, the participants were again debriefed and we discussed the aims of the study with their game experience. Further discussion of the player experience was also conducted online through email correspondences with participants 6, 9, and 12 and WhatsApp communication with participants 1 and 2. The total number of hours of recordings, including think-aloud and interviews, are listed in Table 2 below. Our institution's Institutional Review Board approved the research and we obtained consent from the participants to share their video recordings, which included their voices, for research purposes. This paper adopts a grounded theory approach (Cho & Lee, 2014; cf. Werning, 2017) to induce the types of interactional patterns of the participants in their recordings to derive design principles from these patterns.

2.3. Data coding

The data comprising the interviews, the think-aloud recordings, and online discussions through emails and WhatsApp were transcribed, hand-coded by using a mix of open, descriptive, focused, process, and conceptual coding (Cho & Lee, 2014), and then imported into NVivo to facilitate the analysis. First, using open and descriptive coding, we extracted data from the interview transcripts, laboratory gameplay observations, and participants' reflections and

Table 1. Profiles of the study's participants.

Participant	Age	Gender	Game experience	Game(s) played	Gaming platform	No. of gaming sessions
1	25	M	>15 years	<i>Beyond: Two Souls</i>	PS3	5
2	33	M	>15 years	<i>Beyond: Two Souls, Mass Effect, and The Witcher 2</i>	PS3 & Computer	1, 24, 18
3	21	M	11–15 years	<i>Bioshock</i>	Computer	5
4	23	M	11–15 years	<i>Bioshock</i>	Computer	25
5	23	M	11–15 years	<i>Bioshock</i>	Computer	20
6	19	F	6–10 years	<i>Bioshock and TWD Season 1 Eps. 1–3</i>	Computer	26 and 9
7	22	M	11–15 years	<i>TWD Season 1 Eps. 1–5</i>	Computer	21
8	22	M	11–15 years	<i>TWD Season 1 Eps. 1–5</i>	Computer	20
9	24	M	11–15 years	<i>Mass Effect and The Last of Us</i>	PS3 & Computer	14 and 8
10	26	F	> 15 years	<i>The Last of Us</i>	PS3	8
11	31	M	> 15 years	<i>The Last of Us</i>	PS3	7
12	21	M	11–15 years	<i>The Witcher 2</i>	Computer	11

Table 2. No. of hours of recordings analyzed.

Participant	Selected game	Think-aloud video recordings (hours analyzed)	Interviews (hours analyzed)
1	<i>Beyond: Two Souls</i>	10.2	5.7
2	<i>Beyond: Two Souls</i>	10.5	5.3
2	<i>Mass Effect</i>	30	6.7
2	<i>The Witcher 2</i>	31.9	1
3	<i>Bioshock</i>	9	4.9
4	<i>Bioshock</i>	12.7	5
5	<i>Bioshock</i>	16	5
6	<i>Bioshock</i>	19.5	2.84
6	<i>TWD Season 1 Eps. 1–3</i>	7.2	1.37
7	<i>TWD Season 1 Eps. 1–5</i>	15	6.7
8	<i>TWD Season 1 Eps. 1–5</i>	11.2	6.5
9	<i>Mass Effect</i>	24.4	5.5
9	<i>The Last of Us</i>	23.7	7
10	<i>The Last of Us</i>	14.7	4.5
11	<i>The Last of Us</i>	16.4	6.3
12	<i>The Witcher 2</i>	34.6	0

think-aloud verbalizations in the video recordings to identify the interactional patterns created from their choices and verbalizations. The choices include, for example, selecting dialogue options, interacting with game objects and characters, and choosing whether to participate in the quick-time events (QTEs). The researchers used focused coding to categorize the data. Similar codes were placed into categories. Four broad, higher-order thematic categories were created in this manner: exploring, projecting, playing, and watching the story (cf. Saldana, 2016). For example, visual indicators and verbalizations in the recordings that showed participants selecting all the dialogue options and controlling the character(s) to move around the game world to interact with various objects and other characters in the environment to uncover story information were grouped into “exploring the story.” Visual indicators and verbalizations in the recordings that showed participants selecting dialogue choices and performing actions based on their personality, emotions, and values were grouped into “projecting the story.” Participants’ actions that indicated them making sense of the game controls, formulating strategies, and selecting multiple approaches to attain the objective were grouped into “playing the game,” while those that showed participants not selecting the dialogue options or participating in the QTEs and commenting on the story while watching it unfold onscreen were grouped into “watching the story.”

We argue that the distinction between “exploring the story” and “playing the game” is harder to establish in games where the narrative is very strong and the gameplay is very weak such as *TWD* and *BTS*. In these games, playing the game is closer to exploring the story. The distinction between “exploring the story” and “playing the game” is easier to establish for games such as *TW2*, *TLOU*, *ME*, and *Bioshock* in which both the narrative and gameplay can sometimes be decoupled from each other. For instance, in *TW2*, participant 2 mentioned that there are areas in the game that he can choose to focus on repeatedly killing the enemies to gain experience for his character and obtain money and loot that are dropped by the defeated enemies (Toh, 2021). These actions have no relevance to the story. In *ME*, participant 9 mentioned that the side missions that involved gathering gameplay resources on the different planets did not have any relevance to the story. For *TLOU* and *Bioshock*, there is no narrative involved when the participants were fighting generic enemies when moving from one location in the game world to the next. Coding was conducted until saturation was reached when participants could be observed reporting similar experiences in their qualitative interviews and think-aloud verbalizations and no new information or themes were observed in the data (Guest et al., 2006).

Process coding was used to search for codes that were repeated frequently throughout the data such as participants’ choices and verbalizations that indicated that they were exploring the story to form this thematic category. Finally, conceptual coding was employed to identify salient elements within the data and triangulate the findings with the help of theoretical concepts (e.g., theories on psychological

immersion and interactional patterns). Several strategies such as triangulation, member checking, representative quotations from the interviews and think-aloud, and participant debriefing were used to improve the credibility of the study’s findings (Cho & Lee, 2014). To reduce the incidences of over-interpretation of the data, the codes were regularly rechecked. The coding was checked by a second coder and a consensus was reached about any disagreement in the coding.

2.4. Description and rationale for game choice

Beyond: Two Souls (BTS) is a third-person interactive drama action-adventure game. The player controls the primary character, Jodie Holmes, and a secondary character, Aiden, to participate in the story and learn the life story of Jodie from childhood to adolescence. The *Last of Us (TLOU)* is set in a postapocalyptic world and the player primarily controls Joel, the main character, to explore the story world during and after the postapocalyptic outbreak. *The Walking Dead (TWD)* is set in a fictional world overrun by a zombie apocalypse and the player controls the main character, Lee Everett, who is accompanied by several survivors, including a young girl called Clementine, to endure in the apocalyptic world. *Mass Effect (ME)* is a role-playing third-person shooter where the player controls a human hero, Commander Shephard, to save the galaxy by defeating the main antagonists. *Bioshock* is a role-playing first-person shooter and the player controls Jack, a character who is compelled to defeat the antagonists and liberate the inhabitants from servitude to the city. *The Witcher 2 (TW2)* is a third-person role-playing game and the player controls a monster hunter, the titular Witcher, who must clear his name by unravelling the mysterious identity of the antagonist.

The games were chosen as they were popular commercial narrative games well-known to the participants. They have both prominent story and gameplay elements so that the participants could be studied to understand how they negotiate these elements to progress the game. Finally, a variety of game types were selected to support the study’s aim of understanding the player experience to suggest design principles for improving narrative games.

3. Results

From the analysis of player experiences, we identified the types of interactional patterns of the participants when they engaged with narrative games. These interactional patterns include: (1) playing and exploring the story, (2) playing and projecting the story, (3) playing the game, and (4) watching the story. Using the “items clustered by coding similarity” in the cluster analysis function of NVivo, we selected Jaccard’s coefficient to analyze similarities between the coding categories. We found that “exploring the story” is the highest (overarching) level category, followed by “watching the story,” and finally “playing the game” and “projecting the story” are found on the same lowest (derivative) level of the dendrogram (see Figures 1 and

Items clustered by coding similarity

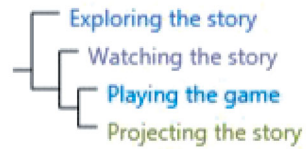


Figure 1. Horizontal dendrogram.

Items clustered by coding similarity

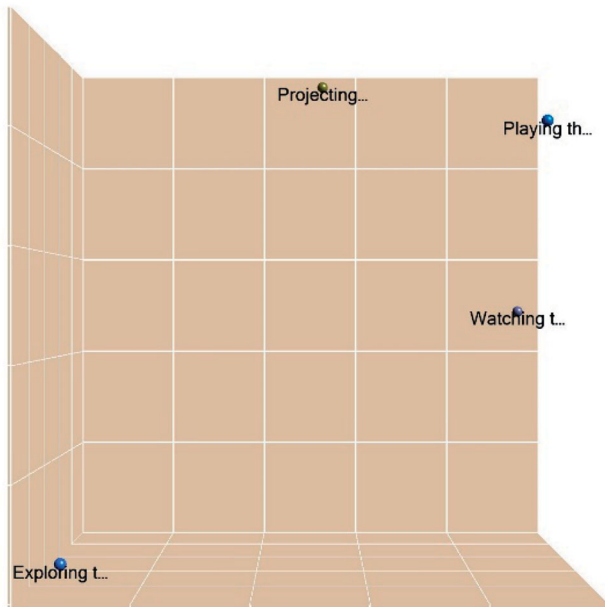


Figure 2. 3D cluster map.

2). As such, the thematic categories include “exploring the story,” “from exploring the story to projecting the story,” “from exploring the story to playing the game,” and “from playing the game to watching (and projecting) the story.” Due to space constraints, only selected representative examples from the data are included for the discussion.

3.1. Exploring the story

The cluster analysis using NVivo indicated that “exploring the story” is the default (overarching) action that was adopted by the participants when they first played the game. The observation of the participants’ playthrough in the laboratory, especially for those playing the first time, and the analysis of their gameplay recordings revealed that they followed the game prompts and instructions closely to move the story forward and explore the story when they started to play the game. This approximates with Van Vught’s concept of the “cooperative player,” who is described as one who plays according to the “predispositions laid down ... by the text” (Iser, 1980, p. 34). However, the participants following prompts initially while they were feeling out the possibility space of the game does not necessarily mean that they are a “cooperative player” following “predispositions” of the text

as their actions can change over time when they preferred to explore the game space on their own or it can be different for participants who preferred to explore the game space on their own.

3.1.1. Exploring TWD, ME, and TW2’s dialogue options

Participants 2, 6, 7, 8, 9, and 12 who played *TWD*, *ME*, and *TW2* were observed to explore all the different dialogue options to understand the outcome of selecting the different options and to deepen their understanding of the story by obtaining the backstories from the NPCs. Participant 2 who played *ME* mentioned that if he did not choose to explore all the conversational dialogues, then he would miss out on the role-playing narrative experience as well as the experience points that he would obtain to strengthen his character in the gameplay:

If I haven’t asked him anything just now, I will be continuing the storyline without having an idea that all the crew members are actually quite on edge because you don’t send so many good people on just a test on a new technology. It will not affect the gameplay a lot if I didn’t ask too much, but I will have missed out all the fun of an RPG if I didn’t do that ... I think you have noticed already, you get experience just by talking to people and you get different experiences by the choices you make.

3.1.2. Exploring Bioshock, TWD, TLOU, ME, and TW2’s environment

Another instance of exploring the story involves the player’s use of the PC to explore the surroundings and interact with objects to uncover more story information to deepen their understanding. Participant 6 who played *Bioshock* and *TWD* highlighted that if she did not explore the different areas in the game, then she would miss out on the embedded narrative such as the audio tapes and part of the storyline as well as gameplay resources such as health kits, weapons, and ammunition. Participants 2, 6, 7, 8, 9, 10, 11, and 12 were observed to explore the game environment to uncover more story information by interacting with objects. As a result of their deeper exploration of the game environment to uncover more background story from the game world, participants 2, 6, 7, 8, 9, 10, 11, and 12’s multiform stories of the games are more elaborated compared to the other participants who focused less on the exploration of the game environment to uncover more background story.

3.1.3. Exploring BTS

During the first interview, participant 1 mentioned that he was not used to playing narrative games like *BTS*, as he preferred to play games for the sake of the gameplay:

Being a gameplay sort of gamer, I find playing this game itself is not very intuitive for a gamer because there’s not much action going on ... There’s a lot of character interaction. There’s this dual gameplay going on between Jodie and Aiden, switching between the two. And for the most part, you don’t know which one is which and you have this sense that you can choose between them, which is interesting. I didn’t play that much as Aiden. I think it’s interesting to explore that idea.

In the first session, when I observed participant 1 playing in the laboratory and talked to him about his game

experience during the interview, he mentioned that when he began playing the game, he was basically just exploring different dialogue options that were provided to see the outcome of selecting the different options:

As for myself, I feel a bit detached from the situation ... I like to do that. It's a personal mechanism ... I'm just sitting here in my chair, you know ... let's see what happens if I do this, let's see what happens if I do that?

During the last interview, participant 2 told me how his experience of playing visual novels influenced how he interacted with *BTS* by exploring the different dialogue options: "For this kind of game, I will just put a save point and go through everything ... [I will try to explore everything] and after that I will just go to the next chapter."

Both participants 1 and 2 who played *BTS* did not engage in a deeper exploration of the story, as they did not know that they could use the PC, Aiden, for further exploration of the story world or perform specific actions due to the lack of explicit prompts. During the last interview, participant 1 explained that it was his first playthrough and there was no prompt to guide him to use Aiden whenever the player could use Aiden to perform specific actions:

I would have been more interested if I had known, you see. The problem is, I didn't know! There's no prompt. There's no suggestion. If I had known that you could use Aiden to do all these fancy things, then I would have, you know, I would have done it more ... Let's be fair, it's a single playthrough, correct?

Consequently, it became evident that participants 1 and 2 tended to project their own version of the story onto the player avatar (cf. Burgess & Jones, 2020) when I elicited their narrative interpretations during the last interview. These participants were more likely to fill in the gaps in their understanding of the game's narrative to make the story more coherent the less thoroughly they explored the story. However, their multiform stories may also be less elaborated because they have not invested time in exploring the story in the game to obtain a deeper understanding of it, but instead filled it with their own projections of the story based on their prior gaming or real-life experiences (Ryan, 2018).

3.2. From exploring the story to projecting the story

This thematic category is induced from NVivo's cluster analysis by coding similarity that groups "exploring the story" and "projecting the story" into different clusters. Projecting the story involves the participants' higher-order cognitive processing and interpreting of the game's narrative to form their personalized stories (Toh, 2018). The cognitive process involves the participants' selection of information from the game world as they explored it, their reflection on the information, and their integration of the information into a coherent whole in the mental model (Toh, 2018; Schröter & Thon, 2014; Upton, 2015) to gain a deeper understanding of the story world along with its rules and mechanics. The integrated information is projected into a personal narrative (Mejeur, 2019). This process may occur during the gameplay and was elicited via the think-aloud verbalizations, during

the interviews in which knowledge was socially negotiated between the researcher and the participant (Girvan & Savage, 2010), or after they have completed the game when they reflected on it in the recordings or communicated to the researcher through email. Projecting the story may also involve the participants' projective identities, values, beliefs, morality, assumptions, and biases (Gee, 2007).

3.2.1. Projecting *BTS's* story

In *BTS*, participants 1 and 2 projected their personality and values onto Jodie when selecting the dialogue options to personalize the story rather than being influenced by the narrative information to make a choice for a specific story outcome, as they mentioned that they did not know what the outcome would be. Both participants discussed how they wanted to create a "good" Jodie, which was like how they perceived their ideal selves to be in real life (Werning, 2017). Both participants chose to save Cole in the chapter "Black Sun." During the last interview, participant 1 explained that it was because he felt that it was the right thing to do:

I think this is a narrative choice, I didn't really think too much about it. I thought it's just [to] heal Cole and get it over with ... Do you think Cole is a bad guy, do you think he shouldn't have to live? That kind of thing. But Cole's a good guy. When you look at it, Cole's a pretty good guy. So ya, doing your best to keep him alive ... I would like to see the story end that way that's why I did it ... I felt like I wanted to push the story that way, so yes, that's why I did it.

By projecting his own personality in playing Jodie in the chapter "The Party," participant 2 chose to leave the house instead of choosing to seek revenge at the birthday party when Jodie was locked in a cupboard underneath a staircase after she revealed Aiden and his powers to the other children. He explained that he would choose to move on if such a scenario occurred in real life and, thus, he left the house. Furthermore, participant 2 revealed during the last interview that when he made the final decision at the end of the story, he was playing Jodie as his player avatar, through which he projected his sense of self to decide what to choose based on his real-life beliefs. Therefore, he chose life by projecting what he would choose acting through Jodie because he felt that he as a human being can die anytime in reality since he has only one life. In contrast, he mentioned that if he were to play Jodie as his PC and select the options based on what he thought Jodie would choose, then he would choose death because Jodie would have been happier by going beyond life.

BTS's participants 1 and 2 also projected their own emergent narrative onto the silent PC, Aiden, during the last interview to personalize it. This is another instance of how the participants' multiform stories differ based on their different interpretations of the game's story as they created multiple parallel possibilities from their projections of their game choices and actions. Participant 2 characterized Aiden as a mischievous entity who they can control to do stuff that Jodie did not want him to do, whereas participant 1 characterized Aiden as Jodie's protector:

In separate segments, yes. [The narrative] gives the gameplay direction. It makes you want to project yourself in that position,

right? If I were Aiden, and I am such and such a character, what would I do? So, the narrative in that sense drives that projection of what you think Aiden is to do the stuff that he does [and characterize] Aiden as a protective and supportive buddy [to Jodie].

3.2.2. Projecting *Bioshock's* story

During participant 3's last interview for *Bioshock*, he reflected on how the game was pushing him more towards performing morally bad actions than morally good actions in the story. There were more instances he recalled in which the game rewarded his morally bad actions such as when he hacked the turrets, the flying drones, the locked safes, and so on for his (character's) own selfish needs according to the philosophy of Objectivism (Murnane, 2018). In contrast, there were lesser instances in which he was rewarded for being selfless such as when he performed the last rites for an NPC by pressing the button to cremate him at the Eternal Flame Crematorium. Additionally, in *Bioshock*, because the game did not explain to the players why the NPCs could communicate with them during gameplay, participant 3 projected his own narrative of how the NPCs such as Atlas could communicate with him through the radio. During the last interview, he assumed that the PC was tracked by a central control system and microchipped so that Atlas could communicate with him. He also assumed that the antagonist NPC, Sander Cohen, found a way to hack the radio communication signal to jam Atlas' communication with the player when his PC arrived at the Fort Frolic level, which was under the control of Sander Cohen.

The personalized narratives of the *Bioshock* participants were created from their exploration of the interactive game space, such as finding, picking up, and listening to the audio logs that they found, stopping to observe and reflect on the visual setting in the game environment, and the subsequent integration and projection of the collected story information into a coherent whole in their mental models (Toh, 2018). When participant 6 explored *Bioshock*, she stopped to observe and reflect on the game environment in the recordings to project her personalized narrative of the game world. On the game level located at Apollo Square, she commented on how the poor people in the underwater city of Rapture were confined by Andrew Ryan's government to living in prison-like conditions.⁴ During the last interview, participant 6 projected her PC's identity by drawing from her knowledge of the game's story synthesized from the audio logs, game environment, and combining it with the *Bioshock* novel that she read for the study (Toh & Kirschner, 2020). She reflected that the in-game resource called plasmids were created by the antagonist, Frank Fontaine and the weapons were created by the underwater city's founder, the main antagonist of the game, Andrew Ryan. She explained that her PC, Jack was like a "union" between the two antagonists in the game, Andrew Ryan and Frank Fontaine. Jack was Andrew Ryan's biological son and Frank Fontaine instructed Brigid Tenenbaum to purchase Jack, as an embryo from Andrew Ryan's mistress, the exotic dancer Jasmine Jolene at Eve's Garden in Fort Frolic. Subsequently, Frank Fontaine

ordered the scientists, Yi Suchong and Brigid Tenenbaum to speed up the growth process in Jack, genetically modify, brainwash, and mind-control Jack to become his "ace in the hole" weapon to take over the underwater city from Andrew Ryan. In the game, Jack was able to use both weapons and plasmids to develop special powers because of the "union" between the two antagonists such as Andrew Ryan who helped to bring the PC, Jack into existence and Frank Fontaine who oversaw his development into a human weapon.

3.2.3. Projecting *TWD's* story

During the interview, participant 6 felt that the game's story was pushing her to portray the PC as a "good" character. The PC, Lee Everett started out in the game's story as a criminal who was convicted for murder after he found the Georgia state senator sleeping with his wife and accidentally killed him in a fight. Participant 6 felt that the PC was meant to do good throughout the game's story to make amends for what he has done. There were instances in the game's story that pushed her in a "good" direction. For instance, she mentioned that the NPC, Lily will make her feel bad when she chose to kill Lily's father, Larry in the meat locker in Episode 2. If she chose to kill the Saint John brothers, Andy or Danny, Clementine will see what she has done and make her feel bad about her choices. At the end of Episode 2, she also felt bad about stealing the provisions from the abandoned car that belonged to the Stranger who appeared in Episode 5. However, she also mentioned that it was ultimately up to the player whether they want to craft a "good" outcome or "bad" outcome for the PC in the game's story (Schrier, 2014, 2017; Sicart, 2009). Personally, she felt that she made the choices in the game based on what she felt was right in the context of the game's story.

When participant 7 explored the story and interacted with the NPC Danny Saint John in *TWD Season 1 Episode 2*, he characterized Danny as a serial killer based on his observation of the NPC's mannerisms. He mentioned that he was creeped out by Danny and his gun when he interacted with Danny at the farm. In the final fight with Danny Saint John, he chose to spare Danny by projecting his ethical values. He reflected on his choice by explaining that:

I was tempted to kill Danny because he was more of a psychopath. Remember how I say he keep looking like a serial killer... The last part when he said that I didn't have the guts to kill him... At that point, I was a bit tempted to kill him, but firstly, that will be wrong. Secondly, if I was Lee, I know shouldn't be drawn in that impulse moment of wanting to kill him. Thirdly, I think the most important one is because Clementine was there, and I didn't want Clementine to see me as a killer.

During the interview, participant 8 mentioned that he was someone who preferred to have a canonical storyline when he played games. His preference made it easier for him to make choices in the game to be in line with the story in the television series, such as in Episode 1 when he chose to support Glenn to leave the group and return to Atlanta to look for his friends after he listened to a radio broadcast

on how bad the situation was in various cities. In contrast, participant 6 mentioned that she chose to support Glenn to leave the group to do what he had to do before he died in the zombie apocalypse because of Glenn's character in the game's story and not because she wanted to make a canonical storyline that was in line with the television series as she felt that the game's story was quite different from the television series.

Participant 8 also mentioned that he drew upon his real-life personal experience to empathize with the NPCs in the game's story and select the dialogue options when he interacted with them. For instance, he mentioned that he was once like the NPC, Molly, who appeared in Episode 4 and was a lone female survivor who preferred to do things on her own rather than work together with others in a group. Therefore, when he encountered her while exploring the story in Episode 4, he could connect with her using his personal experience when selecting the dialogue options to project his personalized narrative of Molly in the game.

3.2.4. Projecting *ME*'s story

After exploring the stories of the various squad members of the PC in *ME* through the conversational interactions, participant 2 achieved a personal understanding of his squad members. He projected their narrative in his reflections in the recordings. He reflected that Ashley had died because of his decision, and he had developed a relationship with Kaidan because of his choices and constant interactions with him on their ship, the Normandy. He could not develop a close relationship with Garrus because he reflected that the "good" female Shepard he was playing was a direct opposite of what Garrus was. He felt that Garrus and Ashley were similar in character to each other, and they were rash, arrogant, and stuck close to their beliefs. He perceived Garrus as a character who believed in justice, but he will do anything, even use illegal means to achieve his objectives, which was not the correct thing to do. He felt that Ashley was xenophobic and believed in humans only, which was very patriotic. However, in *ME*, being xenophobic did not serve any purpose because the whole crew on the PC's ship is made up of aliens. He also mentioned that the storyline told him that Tali was a decryption expert who can help his PC get items in the gameplay, but she also died a lot when he brought her out for mission. He decided on his strategy of using Tali in the gameplay by checking out the game walk-throughs online and by observing that her visual representation of carrying a sniper rifle on her character showed him that she was better suited for range combat. Finally, participant 2 reflected that he decided to find Liara only because he felt that he could form a lesbian relationship with her using his PC, female Shepard.

Participant 9 reflected during the last interview that the exploration of the interactive game space mainly unlocked more story through the conversational dialogues that the PC engaged in with the NPCs, such as his squad members. For instance, by interacting more and more with his squad members, he mentioned that he could unlock individual missions for his squad members such as Garrus and Wrex.

He also mentioned that the romance options in *ME* was quite limited in the sense that the male PC could only form romantic relationships with the NPCs, Ashley or Liara and the female PC could only form romantic relationships with the NPCs, Kaidan or Liara. He reflected that the exploration of the interactive game space did not uncover any narrative artefacts except for Wrex's side mission in which he found Wrex's armour in a locked container. He projected the antagonist, Saren as a typical villain. As he played the game, he uncovered story information through the conversational interactions with the NPCs that Saren was indoctrinated to think that he was trying to save the galaxy and was not the mastermind of the Reaper invasion in *ME*. He also reflected that the main antagonist, Sovereign was interesting only because the player did not understand much about him.

3.2.5. Projecting *TW2*'s story

TW2 was like *ME* in the sense that the exploration of the interactive game space also unlocked more story through the conversational dialogues that the PC engaged in with the NPCs in the game world. When participant 2 completed *TW2*, he reflected that most of his choices were guided by his decision to play his PC as a nice guy⁵ who did not like to kill a lot, which was also what he perceived himself in real-life as. He mentioned that he would stick to that decision when playing other games as well. In *TW2*, he reflected that he did not let the sorceress, Sheala de Tancarville die in the quest "Enter the Dragon" in chapter 3 by choosing to remove the flawed diamond from her megascope to allow her to escape by teleporting away. He also chose not to kill the dragon who was spellbound and controlled by the sorceress, Philippa Eilhart in the quest "Enter the Dragon" in chapter 3. However, he reflected that he could not see the effects of his choices on the narrative (Power et al., 2019) as he did not play the other pathway and both Sheala de Tancarville and the dragon did not appear in the story of *TW2* again after he let them off and their fates were left unknown in *TW2*'s narrative. He also felt that his decision to spare both Sheala de Tancarville and the dragon did not affect his gameplay later as he can still kill soldiers after that.

Participant 2 felt that giving romance options and sexual encounters in the game can motivate the players to continue to explore and play the game. Participant 12 felt that the sexual themes in *TW2* were appropriate (adequately) to a certain degree. He felt that mature themes were necessary to tell the story properly, especially in the world portrayed in the *Witcher*. It showcased the motivations behind the actions of certain characters. Examples of these include King Foltest and the political drama behind his sexual partners, which involved his sister and his secret mistress, the baroness Louisa La Valette, which led to his offspring either being cursed (Adda) or were bastard children respectively. Participant 12 projected his PC, Geralt's personal motivations to be one of lust for women, which he felt was unusual for a *Witcher* as *Witchers* were supposed to be devoid of all sexual weaknesses. His projection of his PC's motivation made him felt that his PC was not a "full, run of the mill"

Witcher, and instead of living strictly to the Witcher's Code, he dabbled in politics and sometimes defended the monsters instead of the humans. For instance, in the quest "With Flickering Heart" in chapter 2, participant 12 chose to side with the succubus instead of the elf as he found the metal shard of a murder weapon that he extracted from the corpse in the catacombs, which he concluded belonged to the elf.⁶ He assumed that the succubus had no need of such weapons to murder another human being or elf and thus he concluded that the elf was the murderer. Participant 12 reflected that without the proper representations of sexual desires in the narrative, it might lead to players seeing the world as a highly innocent one, showing only one side of the characters and making the characters seemed undeveloped.

Participant 12 felt that the representation of the sexual themes in *TW2* did cater to the male fantasy as all he ever saw were overly sexualized women and hardly any sexualization of men in the game (c.f., Heritage, 2020, 2021; Lankoski et al., 2022). Examples include the succubus in the quest "With Flickering Heart" in chapter 2 who was always displayed topless, the prostitutes in the Kaedweni brothel in chapter 2 who were fully naked during the sexual encounters with participant 2's PC,⁷ and the fully naked prostitute interacting with the fully clothed, NPC Dandelion in the vision shown by the prototype megascope in the quest "The Secrets of Loc Muinne" in chapter 3. Sexual encounters were part and parcel of the game and were almost always displayed as a choice for the players to reward themselves when they completed the quests, such as when participant 12 sided with the succubus in "With Flickering Heart." However, participant 12 did not think that was the driving point behind the sexual themes in the game. Instead, he felt that women in the Witcher world, which included the sorceresses/witches in the political organization, the Lodge of Sorceresses, had immense agency, and this is not even seen in our world today. Looking beyond the representation of female characters such as the sadomasochism portrayed by the sorceress, Philippa Eilhart in which she was clearly lesbian serves to drive home the point of sexual liberalism in the world of *TW2*.

Participant 12's projection of the immense agency possessed by the sorceresses and his understanding of their motivations helped him to choose to let the sorceress, Sheala de Tancarville die in the quest "Enter the Dragon" in chapter 3⁸ when she asked him to help her remove the flawed diamond from the megascope. He explained his choice by saying that with the sorceresses, Philippa Eilhart and Sheala de Tancarville as leaders of the Lodge of Sorceresses who did not have the best interests of the north at heart, they will remold the north into their own image if they were to control it. He concluded that it would be better to let her die and totally dismantle the Lodge of Sorceresses. Similarly, he also chose to kill the dragon⁹ to complete the quest "Enter the Dragon" in chapter 3 because he explained that the dragon was under Philippa's control, and the sorceress only had the interests of the Lodge at heart. Since he had allowed Sheala de Tancarville to die, Philippa will want to seek vengeance and so he did not want the dragon to

come after him in the future. Although it will be against the Witcher's code to kill intelligent creatures and against Geralt's psyche to kill the dragon, in this case it was different to participant 12 because he felt that the dragon did not possess free will as she was under Philippa's control.

Like participant 9, participant 12 felt that the romance options in *TW2* was limited. He felt that the romance options were shallow and proceeded in a simple linear fashion that affected one NPC at a time. For instance, the PC can develop romantic relationships, which can eventually develop into sexual encounters with the NPCs in the game's story. Some of these sexual encounters felt meaningless to participant 12. For instance, he commented in the recordings that the writing was not good for the sexual encounter in chapter 3 with the NPC Cynthia, a mage in the Nilfgaardian Empire's service. He felt that there was no depth to the romantic choices when he explored them. Unlike *TLOU*, *ME*, and *TWD* where the PC can constantly interact with their companions throughout the game to develop a relationship and form an emotional connection to them, his PC in *TW2* was only involved with Cynthia in the quest in chapter 3 titled "The Secrets of Loc Muinne." After participant 12 had completed the quest, he can interact with Cynthia by choosing the conversational dialogue options to talk to her. However, there was only 1 line of dialogue which showed Cynthia confessing to his PC that she liked him when participant 12 chose to ask Cynthia about the antagonist, Letho and she advised the PC not to trust Letho as he was a liar. The romance options appearing in the conversational dialogues in the game's story that eventually led to the sexual encounter that were shown to the players in the cutscenes, especially the one that involved Cynthia felt forced to participant 12.¹⁰ He also felt that the writing of the story detailing the developing relationship between his PC and Cynthia as well as the graphics in the cutscenes, such as the one which showed his PC's sexual encounter with Cynthia were not very good. He felt that the story did not allow the PC more time to develop his relationship with Cynthia and the sexual encounter did not integrate into the game's story, leading to a less meaningful projection of the romantic relationship that the PC can form with the NPC in the game's story.

3.2.6. Projecting *TLOU's* story

Participants 9–11's projection of *TLOU's* story was mainly done through exploration, finding narrative artefacts such as handwritten notes and audio recordings in the game world, and the reading and listening to the stories found in these narrative artefacts to project their personalized stories. Participants 9 and 10 reflected that the game was designed to encourage players to explore the environment to find narrative artefacts and gameplay items. Because the stories found in the narrative artefacts were more open to interpretation, participants 9–11 found that it was easier for them to project their own interpretations of the narratives found in these artefacts. Participants 9–11's personalized narratives of the side characters in the narrative artefacts were also different from one another. For instance, during the last

interview, participant 9 projected the side character, Ish as a more positive, hopeful, and light-hearted character who had more inherent goodness in him and who wanted people to come along with him compared to his PC, Joel who preferred to do things alone (Toh, 2018). In contrast, participant 10 projected Ish as a more tragic character who killed all the children that he was with in the community staying in the sewers to stop them from turning into the Infected instead of trying to save everyone from the Infected. Participant 11 felt disappointed that he did not see Ish in the game and wondered what happened to him after he escaped from the sewers.

Participant 9 reflected that because he did not spend time with the side characters who did not appear in the game compared to the main characters who appeared in the game, he could not develop a deeper and stronger emotional connection with the side characters even though he could project the side character's motivations after reading the narrative artefacts left behind by them in the game world. Participants 9 and 10 also reflected that the handwritten notes that they found while exploring the game world helped to contextualize and motivate the gameplay. For instance, when they found the docks note in the chapter "The Quarantine Zone" in *TLOU*, they projected that the note was written by one of the smugglers working under the antagonist, Robert and explained why the smugglers, which they will fight later in the gameplay, were there.

3.3. From exploring the story to playing the game

One of the interactional patterns that has been observed in the data involves the participants' shifting focus from exploring the story to playing the game. This thematic category is induced from NVivo's cluster analysis by coding similarity that groups "exploring the story" and "playing the game" into different clusters. Playing the game can involve the use of gameplay as narrative technique (Sim & Mitchell, 2017) in which the gameplay is used to tell stories in the game. It can also involve the player experience of psychological immersion (Thon, 2008) due to the player's shifting attention to the gameplay (ludic immersion) from the narrative. It may also involve the player reflecting on the gameplay mechanics and strategizing to decide to get the gameplay reward or to enjoy the narrative.

3.3.1. Playing *BTS*

In participant 1's last *BTS* interview he reflected that even though the story is more focused on Jodie, Aiden is always in the background, playing the supporting character. He felt that he missed out on a lot of exploration to collect gameplay bonuses using Aiden, as he did not play the game thoroughly in the laboratory during the first playthrough. He cautioned against the idea of separating gameplay and narrative in *BTS*:

We have to break the illusion that narrative is narrative. Gameplay is gameplay. Because eventually, the time will come when the narrative and the gameplay go hand in hand. So, if you are saying that gameplay decisions affect the narrative, I

would say that the gameplay is a means to impact the narrative. But not in the way that gameplay is distinct from the narrative. Because what these game developers are doing is, they are weaving, they are trying to fuse the two to become one in some games.

Participants 1 and 2 mentioned that there were some instances where the focus in *BTS* was more on the gameplay actions. Participant 1 provided the example of the chapter involving Jodie's training with the CIA that serves as a gameplay tutorial to teach the player how to participate in the QTEs. In the chapter, Jodie hinted to him to use Aiden to distract the soldiers for him to get to the objective, which involved prompts that were supportive of the gameplay. At the obstacle course, participant 1 was given the choice to adopt multiple approaches to knock out the enemies as part of Jodie's training for the CIA:

You know the CIA training that Jodie goes through? Then the final course where she has to knock out some fellows? I'm not sure if everything is 100% linear. Oh ya, there was this scene where you can choose to run and take cover. If I'm not wrong, you can take multiple approaches to knock out the same person? You can choose to take path A or B to knock out the same person. That's gameplay choice. It's clear that it doesn't matter how you knock him out because it's an obstacle course so to speak. It's just different routes. So, it's clear that that one is just the gameplay choice.

Furthermore, participant 2 mentioned that the chapter "The Mission," in which Jodie took on an assignment to assassinate a Somali warlord was more like gameplay because there can be a success (do not get captured) or failure (get captured) state depending on whether the player completed the QTEs or succeeded in sneaking around. However, regardless of the success or failure states, the narrative continues with minor variations.

3.3.2. Playing *Bioshock*

The *Bioshock* participants 3–6 were not focused on the environmental objects such as the photos on the walls, which contributed to the visual storytelling of the story world during gameplay. When these participants collected the optional audio logs, they did not focus on listening to the various NPCs' stories when they were more focused on the gameplay. Participant 3 did not like the game's way of delivering the stories and participant 4 did not fully understand the language used in the audio logs. Participant 4 mentioned that he was not used to the way the characters speak in the audio logs and he could not understand what the characters were communicating due to their accents, slang words, and English pronunciations, so he was unable to relate to the characters. Participant 5 felt that the story given in the audio logs was not relevant to what he was doing in the gameplay. Participant 6 mentioned that the narrative in the audio logs was telling her the story, but the gameplay was distracting her from concentrating on the story. She could focus on the audio logs when she was not fighting any enemies. To achieve a clearer understanding of the narrative in the audio logs, she re-listened to them again after she has finished playing the game.

The various NPCs' stories in the optional audio logs were perceived as noise playing in the background when these participants were more focused on the gameplay to survive, progress the game, and complete the objectives. Most of the time, they picked up the audio logs but ignored their stories during gameplay. Participant 3 explained during the last interview that, "It is the gameplay that is the reason why I continued to play in order to find out more about the story." He preferred that the story be told through the gameplay: "The story and the gameplay are both very flexible such that the gameplay is able to accurately portray the whole story and the story also includes many elements in the gameplay such as having very nice graphics to accurately portray the world of Rapture and the actual depictions of the Little Sisters and the Big Daddies." In this regard, participant 3's multiform story tended to be created through the actions that he performed using his PC in comparison to participant 6 whose multiform story tended to be created through exploring the story.

3.3.3. Playing *TLOU*

TLOU participants 9–11 reflected that the gameplay was sometimes decoupled from the narrative in the game. For instance, there was no narrative when they controlled the PCs to fight the enemies in the gameplay. *TLOU* participants 9–11 reflected during the interviews that the gameplay was used to progress the narrative and whatever they did in the gameplay did not change the narrative and vice versa because of the narrative's linearity. For *TLOU*, the multiform stories of the participants differed more through their projections than from their gameplay actions.

There were other instances in which the gameplay was integrated with the narrative such as when the gameplay allowed the player to participate and immerse in the game's story and experience emotional engagement to the characters. For instance, in the university chapter of *TLOU*, the PC, Joel was impaled on an iron rod and subsequently, the gameplay involved the player to control and bring Joel to safety by following Ellie.¹¹ Participants 9–11 mentioned that the gameplay of Joel moving slowly and having blurred vision was an example where gameplay and narrative cannot be separated as both complemented each other. Participants 9–11 mentioned that when they played this part of the game, they could feel the emotional connection to Ellie when she encouraged the PC, Joel to keep moving so that they can escape from the enemy. When the game involved participants 9–11 in the game and allowed them to play Joel as an injured person with Ellie guiding Joel onwards, they could better form the emotional connection that they felt towards Joel and Ellie. During the last interview, participants 9–11 reflected that playing the game was different compared to watching the events play out in a cutscene since the actions afforded to the players allowed them to better empathize with what Joel was going through in the game's story (Toh & Lim, 2022).

Participant 10 reflected during the last interview that she felt better when she played as Joel compared to Ellie because the game design allowed Joel to carry more weapons

compared to Ellie who did not have many weapons and the player was forced to use a stealth strategy when she played Ellie. Participants 9–11 felt that they could form an emotional connection indirectly through the characters such as Ellie when Ellie talked to their PC, Joel and gave out story information to them gradually throughout the game as the characters were moving from one point to another.

3.3.4. Playing *TW2*

TW2's participant 2 reflected that the gameplay was sometimes decoupled from the narrative in the game. Participant 2 reflected that he was not forced to complete the side quests in *TW2* to develop his character. There were some areas in the game where he could choose to perform repetitive actions to obtain character experience to accelerate the development of his character in the gameplay through grinding. In the game areas such as those in Loc Muinne in chapter 3, enemies including gargoyles could respawn. Participant 2 chose to use the meditation function to shorten the respawn rate for the enemies to repeatedly kill them and increase his character's level in a short amount of time. In this instance, there was no multiform story formed when participant 2 was focused on the actions to develop his PC's gameplay attributes.

In *TW2*, both participants 2 and 12 felt that the gameplay was integrated with the narrative when the player was required to choose between saving or killing an NPC. When deciding which option to choose in the dialogue, participant 2 tended to rely on his personality and what he felt was right to do in the context of the game's story as he was playing the game for the first time. In contrast, participant 12 tended to immerse himself in the PC's psyche and chose the options he thought the PC would choose as he had seen the series and read some of the novels, which provided him with some insight into his PC's character. Both participants could see that their choices have a consequence on the game's story when they read the game walkthroughs online, watched the videos for the game playthroughs on YouTube, or replayed the games after they have completed playing their games. For instance, in the *TW2*'s prologue, participant 2 was required to decide whether to persuade Aryan La Valette to surrender or to fight him to the death. By choosing to persuade the NPC to surrender, participant 2's relationship points with the NPC improved and he was able to meet Aryan again later in the gameplay as they worked together to escape from the dungeon as well as in the narrative in chapter 3 located in Loc Muinne. In contrast, participant 12 decided to fight the NPC, Aryan La Valette in a gameplay event in the prologue and the NPC was removed from the narrative permanently upon his death. When he participated in the prologue's gameplay in the dungeon later, participant 12 met Aryan's mother, the baroness Maria Louisa La Valette who was King Foltest's secret mistress and the Nilfgaardian emissary, Shilard Fitz-Oesterlen instead of Aryan.

Participant 12 also felt that the narrative aspects of the romance and sexual encounters in the game affected the player's choice in choosing to engage in them or not.

For instance, he reflected in the recordings and through email that if he chose to reject the NPC, Triss in the elven baths in the quest “The Rose of Remembrance” in chapter 1, he would get an ability that granted magic resistance as opposed to accepting Triss and getting nothing. His narrative choice then affected his gameplay in a specific way, such as when he chose to immerse in his PC’s psyche to accept Triss in the elven baths and giving up a gameplay advantage to satisfy his narrative decision.¹² Participant 2 also chose to accept Triss in the elven baths, but he did not demonstrate awareness that his choice had any effect on the gameplay as he was playing the game the first time. He mentioned in the recording that his choice had totally no effect on the gameplay except that it provided morale for him to keep on playing the game. Both participants also felt that the gameplay was integrated with the narrative when they had to choose which NPC to side with when playing the game. Participant 2 chose to side with the NPC, Vernon Roche (human) and participant 12 chose to side with Iorveth (elf) at the end of chapter 1, and they continued to different narrative and gameplay pathways that occurred in different places in subsequent chapters of the game. Their PCs met different NPCs and uncovered different story information by completing different quests in the subsequent chapters of the game.

3.3.5. Playing ME

The ME’s participants 2 and 9 reflected during the interviews that the gameplay and narrative are interlinked in this game. Participant 2 mentioned in the interview that the gameplay consisted of the dialogue system in which the PC interacted with the other NPCs, the third-person shooter mechanics, and the gameplay items that the player can pick up while exploring the game world or are given as a reward after talking to other NPCs. The narrative is given in the cutscenes as well as when the NPCs are responding to the PC after the player has selected the choices in the dialogue wheel. ME’s participants 2 and 9 reflected that if they did not talk to the NPCs and completed the main and side missions, then they will not have obtained experience points to add to their character customization that can open up new dialogue options for them to influence the story during character interactions. Participant 2 reflected in the recordings that more than half of the experience points that he obtained for customizing his character and squad members was from completing the main and side missions.

Participant 9 reflected that the character level limitations prevented him from increasing the “Charm” level of his PC over 3 early in the game. When he went to a planet to complete a side mission, he could not select the “Charm” option to save an NPC because of the character level limitation even though he wanted to save the NPC as he was role-playing a “good” PC in the game’s story. During the interview, I also discussed with participant 9 about how completing Wrex’s side mission can help to open up another dialogue option that can be used in lieu of the “Charm” or “Intimidate” dialogue options to calm Wrex down on the mission on Virmire once it is revealed that the antagonist,

Saren has found a cure for the genophage, a biological weapon deployed against Wrex’s people during the Krogan Rebellions. This can be advantageous for players who do not play pure Paragon or Renegade to keep Wrex around as a squad member.

3.3.6. Playing TWD

TWD’s participants 6–8 reflected during the interviews that the gameplay and narrative are interlinked in these games. TWD’s participants 6–8 reflected that those choices requiring them to save, kill, support, or help another character are both narrative and gameplay because making those choices will change the relationship dynamics between their character and the NPCs according to the game rules and mechanics. For instance, when participants 6–8 chose to save Lily’s father, Larry when he suffered a heart attack in the meat locker instead of helping Kenny to kill Larry in *TWD Season 1 Episode 2*, they could see that their choices have an impact on the outcome in the story later when I reviewed the recordings with them during the interviews. They could see that Lily came and help the PC to fight the Saint John brothers later in the game’s story and Kenny just stood there without doing anything to help the PC. Participant 6 also mentioned that she had watched the YouTube recording for the other pathway that shows Lily not helping the PC to fight the Saint John brothers later if she did not help Lily to save her father but helped Kenny to kill Lily’s father. Instead, Kenny came to help her PC fight the Saint John brothers.

During the interview, participants 6–8 mentioned that ethical considerations largely influenced their decisions to save or not to kill the NPCs in the story when they could choose to do so and the NPC, Clementine served as a moral anchor in TWD’s story to influence their choices and actions. Their decisions in the game were also influenced by whether they thought that it made sense for the PC, Lee Everett’s character to behave according to the dialogue options and partly influenced by their own morality of whether it was right to perform certain actions in the context of the game’s story versus trying to survive in the apocalypse. However, participant 6 found it hard to decide when the game forced her to choose amongst equally negative options for the PC to confront the NPC, Kenny after he killed Lily’s father in the meat locker in *TWD Season 1 Episode 2*. It was hard for her to choose because all the options clashed with her personality, which was non-confrontational (Toh, 2021). Like participant 6, participant 8 mentioned during the interview that the choices he made in the game were partly influenced by his personality in real-life. For instance, when he was required to save an NPC or to make decisions prioritizing limited resources amongst the NPCs, he decided based on firstly whether the NPCs were women, children, or family members to the PC, then deciding based on whether the NPCs were friends to the PC, and finally deciding based on whether the NPCs were useful to the PC (Toh, 2021).

Participant 7 reflected during the interview that the scripted narrative events involved him in the story by having

him play as the PC, Lee Everett. For instance, in *TWD Season 1 Episode 1*, he was given the prompt to click to kill the PC's own brother who had turned into a zombie outside the pharmacy to get the pharmacy keys from him. He reflected that the action afforded to him by the game enabled him to connect emotionally with his PC. He could immerse in the story and empathize with the PC's emotions of sadness when he selected the dialogue option to speak with his brother one last time before he selected the prompt to have the PC chop his brother's head with an axe (Toh, 2018). In contrast, participant 8 felt annoyed rather than sad when he played this part. During the interview, he reflected on the relationship between the PC and his brother, such as how close they were with each other before the apocalypse as there was not much information given to the player about the relationship between the PC and his brother in the game (Toh, 2018).

3.4. From playing the game to watching (and projecting) the story

The last interactional pattern observed in the data involves the participants' shifting focus from playing the game to watching the story. The game uses cinematic devices such as cutscenes, scripted narrative events, or QTEs that minimally involve the players in the game's story. It can also involve the player experience of psychological immersion (Thon, 2008) due to the player's shifting attention to the narrative. This thematic category is induced from NVivo's cluster analysis by coding similarity that groups "playing the game" and "watching the story" into different clusters.

NVivo's cluster analysis by coding similarity, which also groups "watching the story" and "projecting the story" into different clusters, indicates how "watching the story" is related to "projecting the story" when the participants could focus on the interpretation of the game's narrative to form their personalized multiform stories as they watched it. Participants 3–12 who played *Bioshock*, *TLOU*, *ME*, *TW2*, and *TWD* highlighted that it was important for crucial narrative information to be given in the cutscenes, scripted narrative events, or QTEs that involved minimal gameplay interactions rather than it being given to them when they were playing the game.

3.4.1. Watching (and projecting) *BTS*' story

BTS seems to fit into what Jesper Juul terms as casual games, which are easy-to-learn family games that tend to use simple technology and can appeal to a broad range of audiences (Juul, 2010). *BTS* also has four out of the five components that characterize casual games. These include usability, interruptibility, low level of difficulty and no punishments for failing, and redundant positive feedback for every successful action that the player performs (Juul, 2010). In the last interview, participant 1 commented about his player experience:

It has a very water-downed and simplified mechanic of gameplay but it's sufficient for this game... They managed to

condense it to this form which captures the essence of, if you see games nowadays, they are getting simpler. The mechanics are becoming more casual. In terms of interactivity, it is just enough.

Both participants 1 and 2 reiterated that the gameplay interactions in *BTS* could be optional during the last interview. Participant 1 mentioned that:

The gameplay interaction is minimal... If you want to say the gameplay drives the story forward, right? The narrative event could have happened without the need for the gameplay, right? ... It could have been entirely a cutscene... Most of the choice you get is the order in which you knock the stuff around... It's all just interactions leading to the same outcome, right? ... They are making use of the gameplay to push forward the narrative.

BTS co-opts elements of idle or self-playing games that tend to play themselves (Fizek, 2018). In the case of *BTS*, the interactive media do not only invite the observer to participate productively in their reception and take over parts of the artistic effort, but the interpassive media can also take the effort of participation away. For example, participant 1 was occasionally observed to let the game randomly chose one of the dialogue options for him during conversational interactions. Similarly, in the latter half of *BTS*, participant 2 decided to let the game run on its own whenever he could choose to do so and did not participate after he realized that the PC will not die in the QTEs (Toh, 2018). This freed the participants to focus on the narrative and project their multiform stories instead of forcing them to press the buttons or select a dialogue option if they did not feel like participating. Consequently, the players were able to experience pleasure from watching the interactive story passively, depending on their player style, mood, and available cognitive resources.

3.4.2. Watching (and projecting) *Bioshock*'s story

Participant 3 preferred that the audio log's narrative in *Bioshock* be given in the cutscenes. He mentioned that the cutscenes did not interrupt his gameplay and they engage the player by giving the player the background of what is happening instead of allowing the player to jump into the game without any contextual knowledge of the game's story. Participant 4 also preferred that the audio log's narrative in *Bioshock* be given in the cutscenes so that he can focus on watching and understanding the story when it was given to him. Even though he mentioned that he had picked up the audio logs when he played the game, he did not focus on listening to them when he focused more on the gameplay. He explained that he did not understand the language used in the audio logs when the characters used slang words and spoke in an accent.

Participant 5 mentioned that when he played *Bioshock*, he focused on the visual part of the game and backgrounded the audio log's narrative. He preferred watching the narrative in the cutscenes, which not only contextualized the gameplay, but also provided him with clues as to how to fight the enemies in the game world, such as the heavily armoured boss called Big Daddy. Participant 6 mentioned that the cutscenes provided the background to the game's

story and the audio logs build on the main story by providing the side characters' stories in the game, which were optional. Participant 6 also mentioned that every area of the game had a meaning to it, and it was the visuals that enriched her player experience. The game used a variety of storytelling devices in the visuals such as the ghosts¹³ to show the story to her and she was engaged by this method of storytelling, which allowed her to watch and project the story of Jasmine Jolene¹⁴ and her relationship to the antagonists, Andrew Ryan and Frank Fontaine. When her PC met Andrew Ryan who explained to her about the controlling phrase, "Would you kindly?" for her PC in a cutscene, she thought that this part of the game's narrative was well-designed. She understood that the game designer, Ken Levine, used the PC's lack of control in a cutscene to explain to her that her PC was being controlled by the antagonist, Frank Fontaine and by extension, undermining the sense of agency offered by the game to the player (Stang, 2019).

3.4.3. *Watching (and projecting) TLOU's story*

Participant 9 mentioned that the story did not interrupt his gameplay when it was given to him either in the cutscenes or by the NPCs when there were no enemies around and he was exploring the area. This allowed him to focus on watching and projecting his understanding of the story. He felt that the cutscenes developed the game's story and characters and allowed him to form an emotional connection to the NPCs, such as Tess who was a main character and he projected her as a strong female character who was an equal to Joel. He felt sad when Tess died early in the game when she was killed by the military in a cutscene. The cutscenes motivated participants 9–11 to continue to play the game and find out what happened next in the game's story.

Participant 9 shared an instance when there was a transition from the gameplay QTE to a narrative event in the cutscene that he felt was well-designed in the game as it created an emotional connection for him with the characters by involving him in the game's story. For instance, in the university chapter of *TLOU*, participants 9–11 were required to participate in the gameplay and control Joel to fight an enemy by pressing the correct button on the PS3 controller in a QTE. Upon successful completion of the QTE, they could see that the gameplay event shifted to a narrative event that occurred in a cutscene to show them that Joel was impaled on an iron rod after falling down a ledge while fighting an enemy. Participants 9–11 watched the narrative in the cutscene and projected the narrative event of Joel being impaled on the iron rod as an important plot point with narrative consequences in the next chapter in which the game allowed them to play as Ellie to take care of Joel in a role reversal to the earlier chapters in which Ellie was dependent on Joel.

Participant 10 mentioned that it would be very distracting if the narrative was given to her simultaneously as she was wandering around and exploring the game world in *TLOU*. She shared that her prior gaming experience was one which was gameplay focused as she frequently played action games and this experience caused her to focus more on the

gameplay and background the narrative elements when she played the game. She did not pay attention to what was being said by the characters or read what was being written in the notes when she performed gameplay actions, such as fighting the enemies or scavenging for items. When the story was given in the collectibles such as the documents in the game, she might collect them, but she might not spend the time to read all the details because she was more focused on the gameplay. The cutscene therefore forced her to pay attention to the narrative, understand, and interpret (project) it. During the last interview, she mentioned that the most emotional scene in the game for her did not occur in the gameplay, but occurred in the cutscene after Ellie killed the antagonist, David in the winter chapter. Participant 11 was appreciative of how the narrative that was given in the cutscenes did not intrude into his gameplay so he could focus on understanding and interpreting (projecting) the narrative when it was shown to him. During the last interview, participant 11 also mentioned that QTEs in the cutscenes were not meaningful to him because he was just following the prompts on the screen, and he did not have any agency or control in what he was doing.

3.4.4. *Watching (and projecting) ME's story*

Participant 2 mentioned during the interview that the cutscenes are important because they are *ME's* way of telling the story and serves to instruct the player what to do to progress the game's story. Participant 2 mentioned during the interview that he was curious to find out what happened next in the story after watching the cutscenes. Participant 9 liked that the game's story was given to him in the cutscenes as he could focus on watching and projecting the story. During his reflections in the recording, he mentioned that cutscenes provided him with a nice break in between the gameplay and the story pacing is good as the player is being fed with the story in the cutscenes gradually. Participant 2 mentioned that the cutscenes motivated him to continue to play the game to find out more about the story and how the story continued. For instance, he mentioned during the interview that when the cutscene first introduced the antagonist, Saren to him, he did not know what he was doing with the beacon on Eden Prime. However, he assumed that the antagonist was up to no good and his first instinct was to kill the antagonist.

During the interview, participant 9 mentioned that the cutscenes were showing him the story from the multiple perspectives of different characters. For instance, in the beginning of the game, he mentioned that the cutscene that showed him the antagonist, Saren killing the NPC, Nihlus was from the perspective of a smuggler who was hiding behind the crates and saw what happened. Then, he interpreted the cutscene that showed him that the antagonist, Saren who felt angry on his ship after he left the beacon on Eden Prime was from Saren's perspective. Besides the cutscenes, participants 2 and 9 also watched the characters talking to each other after they have selected the dialogue options to project *ME's* story. Both participants played the game, selected the Paragon options in the dialogue wheel,

and watched the character interactions. They projected their PC as a good character in the game's story.

There were other visuals in the environment that allowed the players to project the story. For instance, participant 9 liked that the game designers integrated the Conduit, a mass relay device into the narrative setting at the beginning of the game and he projected it as a very good plot device¹⁵ that can transport the PC and his squad members back to the Citadel for the final confrontation with the antagonists at the end of the game.

3.4.5. Watching (and projecting) TWD's story

Participant 6 mentioned during the interview that it was not the immersion in the gameplay but the emotional attachment to the characters and the story that attracted her to play, watch, and project TWD's story. Participants 6–8 reflected that the gameplay was very minimal and consisted of QTEs, point-and-click actions, character movements, and conversational interactions with the NPCs. Participant 6's experience of the game was like watching a drama which was interactive. Both participants 6 and 8 felt that the timed dialogue options made the game's story more realistic by simulating how real-life conversations occur. Participant 6 mentioned that the choices that the player made such as supporting, saving, or helping the NPCs did not have any effect on changing the overarching narrative of the game. Instead, they contributed to minor variations within the game's story, such as affecting the relationships between the PC and the NPCs, which characters will accompany the PC till the end, some minor changes in the story settings, and dialogue variations. However, the minor variations will converge at the end of the game, and there is only 1 single narrative ending.

Participant 7 mentioned during the interview that the cutscenes were important for the player to understand the storyline. The cutscenes were also important because some of the cutscenes not only provided instructions to the player by telling them what to do to progress the game's story by giving the player the general goal and overview of the situation they face in the game, but also allowing the player to find out by themselves how to progress the game segment. He mentioned that QTEs and scripted narrative events that were integrated into the game's story allowed him to play, watch, and project the game's story without being distracted by irrelevant stimuli in the game. He mentioned that the QTEs and scripted narrative events were able to immerse him in the PC's role in the story.

During the interview, participant 8 mentioned that QTEs were quite old mechanics, and they were quite common in video games in the 1990s. He felt that QTEs do not work in any games today because we have more technology, game mechanics, and innovations that game designers can put into games. He also felt that the mainstream commercial games do not have much innovation compared to indie games. He commented that QTEs can be perceived as a lazy way of delivering the story if they were implemented in today's games. Having said this, he still felt that he could immerse in the PC's role when he played the game because

he felt that the integration between the actions that he could perform as the PC and the story is well designed, and the use of QTEs has been established in graphic adventure games such as TWD game from Telltale games.

3.4.6. Watching (and projecting) TW2's story

The cutscenes in TW2 were important for the player to project their understanding of the narrative, and to understand how the NPCs are represented in the game's story to help them to make a choice in the dialogue system. For instance, participant 2 played chapter 2 and watched the cutscenes, such as the one which showed King Henselt betraying his PC by having all of Vernon Roche's men hanged except for Ves even though his PC had helped the king to remove his curse earlier on in the game¹⁶ and he also watched (and later projected¹⁷) the other cutscene that implied that King Henselt raped the NPC, Ves.¹⁸ By watching the cutscenes, participant 2 projected the characterization of King Henselt as someone who was an arrogant person full of himself and amoral.¹⁹ He reflected in the recordings that he was annoyed by King Henselt²⁰ and his projection of the king helped him to choose the dialogue option to let the NPC, Vernon Roche kill the king after his PC had killed the king's guards. Other than the cutscenes, the QTEs were also used as a storytelling device in the game by letting the player participate in the story. However, participant 2 commented in the recordings that he did not like the QTEs, such as those that involved him in the game when he was required to press the correct buttons to fight the two guards to save his companions, the NPCs, Dandelion and Zoltan from being hanged in the town of Flotsam in chapter 1 of the game.²¹ He felt that the action of pressing buttons during the QTEs were artificial to him and broke his immersion in the game's narrative (Qin et al., 2009) when he played the game, watched, and projected the story.

Participant 12 commented in the recordings that some of the NPCs' stories were not very developed in the story of chapter 2 that branched out in two different directions depending on whether the player decided to side with the NPCs, Vernon Roche or Iorveth at the end of chapter 1. After he watched the cutscene that showed him that the NPC, Dethmold was sentenced to be executed after King Henselt's defeat at Vergen, he commented that some of the characters' stories such as Dethmold and King Henselt's stories were not very developed in Iorveth's pathway compared to their more in-depth stories in Vernon Roche's pathway.²² He also commented that Vernon Roche and Ves' stories as well as the sorceress, Sabrina Glevissig's stories were more in-depth in Roche's pathway. The implication of his comments is that if the player were to complete only 1 of the pathways in chapter 2 or if they started with Iorveth's pathway on their first playthrough, their understanding and projection of the game's story will be incomplete, and they will conclude that the story is not very nice.

Table 3 summarizes the four types of interactional patterns that were induced from the study of the player experience.

Table 3. Interactional patterns.

Interactional patterns	Definition	Examples
Exploring the story	The participants explored the different game areas and interacted with objects and dialogue options to investigate their functions and outcomes and uncover story information.	Participants 1, 2, 6, 7, 8, 9, and 12 explored the dialogue options in <i>BTS</i> , <i>TWD</i> , <i>ME</i> , and <i>TW2</i> to understand the outcome of selecting the different options and deepen their understanding of the story. Participants 2, 6, 7, 8, 9, 10, 11, and 12 who played <i>Bioshock</i> , <i>TWD</i> , <i>TLOU</i> , <i>ME</i> , and <i>TW2</i> explored the different areas in the game so that they would not miss out on the embedded narrative and the storyline.
Projecting the story	The participants interacted with the PC as an avatar and formed their own narrative interpretations based on their personality, emotions, and values.	In <i>BTS</i> , participant 1 projected Aiden as a protector of Jodie whereas participant 2 projected Aiden as a mischievous entity. In <i>TWD</i> , participant 7 chose to spare Danny Saint John because of his own ethical values.
Playing the game	The participants focused on the actions, strategies, controls, or objectives instead of the story during the playthrough.	<i>Bioshock</i> participants 3, 4, and 5 perceived the optional audio logs as noise in the background when they focused on the gameplay to survive, progress the game, and complete the objectives.
Watching the story	The participants did not interact with the dialogue options or QTEs but allowed the game to run by itself. It also refers to the participants' focus on watching the story in the cutscenes, scripted narrative events, or QTEs.	Participants 1 and 2 let <i>BTS</i> select one of the dialogue options for them. Participant 2 let the QTEs run by themselves in the latter half of <i>BTS</i> . All participants watched (and commented on) the story in the cutscenes, scripted narrative events, or QTEs in each game.

4. Design principles for narrative games

Based on the study of the player experience, we suggest seven design principles that game designers can consider when designing narrative games to better engage the players and promote their exploration of the story. These principles include player-avatar dynamic, a strong interactive space for the player to explore, placeholder for projecting an emergent narrative, inclusive game design to enhance accessibility, using an interactive feedback loop to adapt to the players' playstyle, QTEs and emotional connection to the characters, and more complex options to romance/relationships in narrative games.

4.1. Player-avatar dynamic

From the findings, we have shown that there is the interactional pattern that shifts from "exploring the story" to "projecting the story" in Section 3.2, the interactional pattern that shifts from "exploring the story" to "playing the game" in Section 3.3, and "playing the game to watching (and projecting) the story" in Section 3.4. Based on these three interactional patterns, the discussion in Sections 3.2.5, 3.3.4, and 3.4.6 has shown that participant 2 projected his personality and feelings onto the PC when making choices in *TW2*, whereas the discussion in Sections 3.2.5 and 3.3.4 shows participant 12 immersing in the PC's psyche to make choices in *TW2*. In Section 3.3.6, we have also found participants 6–8 both immersing in the PC's character role and projecting their morality when making choices in *TWD*. From these findings, we suggest that game designers can create a PC that allows the player to strike a balance between role-playing as the PC and for the player to project their values, beliefs, morality, assumptions, and biases (Gee, 2007) when playing the game (c.f., Grace, 2020). During the final interview with participant 6 who played *Bioshock*, she

mentioned that she could not empathize with her PC who was completely silent, and she did not know much about him until further into the game. Therefore, it is important for game designers to create a PC who the player can not only play as but also relate to. Previous studies (e.g., Huang et al., 2015; Wan & Chiou, 2006) have proven that the main motivation in playing video games might be an innate desire to satisfy an interpersonal, humanistic need. When the player can project their values while role-playing such that there is a balance between playing the character and how the player can behave according to their real-life values, beliefs, and personality, the PC can serve as a very strong extension of the players' psyche, and the player can immerse in the game world.

We can demonstrate the principle of the player-avatar dynamic using the prologue of *TLOU*. In the prologue, the player is allowed to control the PC's daughter, Sarah, before they are given control of the PC, Joel. While playing as Sarah, the player might project emotions of insecurity, fear, and increasing emotional tension as they watched the apocalypse unfold before Sarah's eyes while they controlled Sarah to look for Joel in their house. As the game shifted control to Joel for the player, they can then empathize with Joel's intense need to protect Sarah as they role-play as Sarah's father, watch the cutscenes, and project the important value of family and protecting the family. The *TLOU*'s prologue's narrative build-up is well-designed as the player is gradually brought into the story as they learn the controls gradually while they play as Sarah and then Joel. They can understand why the PC, Joel wanted to protect his daughter by projecting the humanistic value of protecting one's loved ones during the apocalyptic outbreak, and the player is also given the control to bring Sarah to safety.

The narrative build-up occurs in a progressive manner and the player soon felt the increasing danger as they encountered more Infected the closer they reached the safety

checkpoint. However, just as Joel was about to reach safety with Sarah, the player's control of the game was taken away. In a cutscene, Sarah was shot and killed by the military stationed at the checkpoint under orders to shoot anyone who tried to cross the checkpoint. Sarah died in Joel's arms at the moment that the player thought that they have successfully brought Sarah to reach safety through their actions. By role-playing as Sarah and Joel during *TLOU*'s prologue, the player can understand why Joel behaved in a cold manner throughout the rest of the game by playing the game and then watching (and projecting) the traumatic event he experienced in the prologue from his perspective. Essentially, *TLOU*'s prologue provides crucial story information to the player that Joel could not move on 20 years after Sarah's death, which changed him into a ruthless survivor.

4.2. A strong interactive space for the player to explore

The findings of the interactional pattern that shifts from "exploring the story" to "projecting the story" in Section 3.2 and the interactional pattern that shifts from "exploring the story" to "playing the game" in Section 3.3 are also applicable to the principle of a strong interactive space in this section. A strong interactive space allows players to explore the game world and interact with the game objects and characters found in it to come up with their personalized narratives, such as when participant 2 projected his personalized narratives of his squad members in *ME* in Section 3.2.4 to evaluate the characters and beliefs of his squad members. In Sections 3.2.2 and 3.2.3, we have discussed how participants 3 and 6 reflected on their gameplay and projected morals such as their own principals about whether their actions in the game's story were right or wrong. Game designers can consider creating a strong interactive space by presenting game events with ethical issues that the player can interact with such as game objects or characters when they explore the interactive space. Ethical issues that are provided in a strong interactive space can provide the potential for the players to reflect on their gameplay actions. The players can attempt to connect their actions to the game's narrative, but they are also provided the freedom to project their personalized interpretations of the narrative along with their values to become engaged with the happenings in the game world.

We can demonstrate the principle of a strong interactive space for the player to explore using *TW2*'s prologue. In *TW2*'s prologue, participant 2 chose to explore the civilian's homes when he attempted to find the underground sewer system to reach the solar of a temple within the fortress where King Foltest's bastard children were being held hostage. The player controlled the PC to explore the game space and encountered the game events in *TW2*'s prologue. When he played the game, he could project a personalized narrative of how the civilians were usually the victims during wartime and how the aggressors can take advantage of the civilians by justifying their actions to kill civilians. His feelings of injustice for the civilians caused him to choose to stop a commander who was about to set fire to their homes while exploring the interactive game space to complete the

game objective. When participant 2 explored the civilian's homes, he found that he can choose to loot their houses. Although the player felt that the action of looting was ethically wrong, participant 2 projected his personalized narrative that his PC, the Witcher needed to survive in the gameplay, so he chose to take the civilian's items from their houses. He also demonstrated understanding about the game structure by reflecting that the action of looting in *TW2* is separated from the narrative and has no ethical implications or effect in changing the game's narrative.

Game designers can also consider creating a strong interactive space that consists of rich visuals and narrative artefacts in the game environment that can help to bring out the believability aspect of the game's narrative for players to project their personalized narratives of the game world. When the players explore the game space, they can immerse in the visuals, and reflect on the game world to project their personalized narratives by drawing on their prior experiences in the real world. In Section 3.2.2, we have discussed how participant 6 explored *Bioshock*'s game environment and reflected on the prison-like interactive space in the game level located in Apollo Square. She projected her personalized narrative by commenting how Andrew Ryan's government was oppressing the poor people who were living in the slums in Apollo Square. Participant 9 explored *TLOU*'s rich environment and found the audio tapes, notes, journals, and children's drawings left behind by the previous inhabitants who hid in the sewers away from the Infected in Pittsburgh, Pennsylvania. He reflected on the narrative artefacts and the visual representations of the interactive space in the sewers and projected his personalized narrative of how the inhabitants managed to survive living underground in the sewers as a community. However, when participant 9 continued to read the notes left behind by the inhabitants, his personalized narrative was expanded as he discovered that the inhabitants staying in the sewers were killed when the Infected discovered them and managed to enter their camp through an unsecured open door and massacred the survivors except for a few who escaped.

Game designers can create a strong interactive space that requires the players to solve a puzzle or piece together a mystery to uncover the game's narrative. As the players explore the interactive game space, they can discover narrative artefacts gradually that serve to arouse their emotions when they explore and play the game. Participant 6–8's exploration of the Saint John's dairy farm in *TWD Season 1 Episode 2* led them to have a deeper understanding of the story when they found and interacted with the NPCs and narrative artefacts scattered throughout the interactive space. When participant 7 found the tools used by the Saint John family to trap and cut up humans for their meat behind a locked door in the Saint John's dairy farm, he projected his moral values in the story world of the game and expressed his disgust towards the Saint John family's cannibalistic behaviour, such as killing human beings to eat their meat. Similarly, as participants 7–8²³ explored Crawford in *TWD Season 1 Episode 4*, they found out gradually that it was human corruption and draconian rules²⁴ that led to the

downfall of the community. It started to fall apart from within and was eventually penetrated and overrun by the zombies from without. When game designers create the game environment such that players can explore it, talk to the NPCs, and examine the narrative artefacts that hold clues to uncover what was truly happening in the interactive space, the players can experience feelings of curiosity, suspense, and satisfaction as they gradually discover the identities and motivations of various characters in the game.

4.3. Placeholder for projecting an emergent narrative

The findings of the interactional pattern that shifts from “exploring the story” to “projecting the story” in Section 3.2 can be used to derive another game design principle in this Section. In Section 3.2.1, we have discussed how the use of a secondary PC such as Aiden in *BTS* can enable the players to project their values and preferences of what they want the PC to be like in the story. Similarly, in Section 3.2.3, we have discussed how NPCs such as Danny Saint John in *TWD* can be used by the player to project ethical values from real life onto the PC. Therefore, we suggest that when designing a narrative game, the game designers could consider integrating a placeholder character with indeterminacies (Farca, 2018) to enable the players to project their personality, emotions, and values (Griebel, 2006) onto the character so that they can feel a greater ownership, control, and connection to the story by developing their own emergent narratives and multiform stories. Using such a character for player projection could also help to broaden the game’s appeal to the players by allowing them to create their own multiform stories. Players could then personalize the game according to their preferred ways of playing the story using a placeholder character for expressing their personality, emotions, and values.

4.4. Inclusive game design to enhance accessibility

In Section 3.1.3, our findings indicated that the participants could lack awareness that they can use PCs such as Aiden in *BTS* to perform specific actions when there were no explicit prompts to guide the player to interact with narrative artefacts in the game world to trigger a cutscene for them to watch and project their multiform stories. Therefore, we propose that game designers could consider integrating clearer multimodal cues such as those found in *TLOU* to prompt the player to use secondary PCs (or objects), whenever possible, to perform specific actions (or to interact with). Game designers can employ good accessibility design for disabilities²⁵ to deliver information to the player via multiple senses. For example, the secondary character can attract the players’ attention by using a high-pitched sound and its colour can change to alert the player. Such a game design may not only benefit nondisabled players, but it can also be sufficiently inclusive to benefit players with specific impairments (Heron, 2012). One of the recently released games integrating inclusive features that are user friendly for players with impairments include *TLOU2*, which has over

60 different accessibility settings that enable an unprecedented level of customization and fine-tuning to fit diverse needs (McAloon, 2020).

As we have also discussed in Sections 3.3.2, 3.4.3, and 3.4.6 that the player may not focus on the narrative when they are immersed in the gameplay, game designers should consider not overwhelming the players with information from multiple modalities such as language, audio, music, and visuals during gameplay. Crucial narrative information should be given to the player when there is minimal gameplay so that they can focus on watching and projecting the game’s narrative.

4.5. Interactive feedback loop to adapt to players’ playstyle

Based on our findings of the participants’ interactional patterns, including exploring the story in Section 3.1, projecting the story in Section 3.2, playing the game in Section 3.3, and watching the story in Section 3.4, we suggest that interactive stories can be designed to adapt to the players’ degree of interactivity or passivity when playing the game to enhance the player experience (c.f., Paraschos & Koulouriotis, 2022). El-Naser (2007) presents a new interactive narrative architecture designed using a set of dramatic techniques that he formulated based on several years of training in film and theatre. The system can infer a user’s character intention or goals given their actions and story context. Based on user action, the system can use a set of rules to calculate a new value for the narrative vector based on the story context, user action, and previous vector values. A similar system can be implemented in IDN to change the degree of story interactivity based on the amount of a player’s active or passive interaction to facilitate player agency (Tanenbaum & Tanenbaum, 2010) and to personalize the multiform stories to the players. For instance, when the system detects that the player prefers to watch the story unfold through character interactions, the system could provide more compelling dramatic experiences to provide the player opportunities to influence the character relationships meaningfully through specific choices without feeling that the choice was made for them or was too easy to be made (Joyce, 2016; Toh, 2021).

In contrast, when the system detects that the player prefers to perform actions to progress the narrative, it could provide more action-based game scenarios for the player to immerse in the gameplay by giving them more options to participate in the action. However, it is also possible that if a player were too active, they would miss dramatic content. In this case, game designers can consider including an option for players to turn on to explicitly inform them of the effects that their play choices have on the type of content they would encounter. This would be like how *TWD* has an option that can be turned on by players to inform them when their choices have an effect on the relationship between the PC and the NPCs. Game designers could also create a game system that can detect whether and when players need multimodal cues (see Section 4.4) and modify the user interface accordingly to enhance their user experience.

4.6. QTEs and emotional connection to the characters

The findings in Section 3.4.3 have discussed how QTEs can allow participants 9–11 to form an emotional connection to the characters in *TLOU* and the findings in Section 3.4.6 have discussed the example when QTEs cannot allow participant 2 to form an emotional connection to the characters in *TW2*. Following from the findings, we propose that QTEs that are used to involve the players in the game's story will work more effectively if they allow players to connect emotionally to the characters in the story. The *TLOU*'s participants 9–11 could form an emotional connection to the characters because they could participate in the game's story and see and hear Ellie helping Joel as they controlled the PC, Joel to follow Ellie to safety. They could experience the PC's struggle with the enemy in a QTE which caused him to be impaled on the iron rod and became dependent on Ellie. In the example from *TW2*, participant 2 could not develop the emotional connection to his PC and the other NPCs when he was required to press a series of correct buttons to rescue his companions in a QTE that simulates a fighting sequence. He did not feel that the gameplay actions that he could enact in the QTE were meaningful, and they felt artificial to him.

QTEs might not work well if they were merely integrated into the gameplay fights with enemies, but did not connect meaningfully with the game's story. In the case of *TW2*, the QTE that was implemented did not allow participant 2 to experience any "loss" or major plot development when he played the PC, Geralt to save his companions when compared to the example in *TLOU* when participants 9–11 could experience how their PC suffered a major injury in the QTE and became dependent on Ellie in the subsequent gameplay. Game designers should consider integrating QTEs sparingly into the game's story such that they could allow players to form emotional connections to the characters by involving them in the story to experience the PC's struggle or "losses." When the player is allowed to play through QTEs that are integrated into the game's story, they could experience an emotional connection to the characters when they can empathize with the characters.

4.7. More complex options to romance/relationships in narrative games

The findings in Sections 3.2.4 and 3.2.5 have shown that there are limited story options for developing relationships with the NPCs in the game's story. Current narrative games integrate a simple romance system in which there is a linear progression for advancing romantic relationships with other characters in the narrative. Game designers can consider integrating more unpredictable and complex romance options with different NPCs in the narrative. They can think of how multiple relationships can be integrated into the narrative design that complicates the romantic relationships between the PC and other NPCs in the game. Instead of rushing through the romantic relationships to result in a sexual encounter in a short span of a few sequences of dialogue choices, the story can provide more time for romantic relationships to develop between the PC and the NPCs. For

instance, they can allow the player to choose specific NPCs to bring out with them and provide opportunities for the PC to work together with their companions to complete quests or loyalty missions like those found in the *ME* video-game series for a romantic relationship to eventually form.

The romance dynamics can also be made more complicated such that the PC's relationship with the NPC can change from the interaction with multiple NPCs. For instance, in the *Witcher* videogame series, choices that the player can select to improve their romantic relationship with a specific NPC, such as Triss can result in a deterioration in the PC's romantic relationship with another NPC, such as Yennefer when she felt jealous that the PC was with another NPC. More complex relationship dynamics can improve the player experience in narrative games and immerse them in the game's story when exploring and playing the game.

Game designers can consider broadening relationship dynamics beyond romance to include parent/caretaker-child relationships, friendships, sibling relationships, lesbian, gay, bisexual, and transgender (LGBT) relationships, and so on (see Grace, 2020). For instance, Grace (2020) has discussed how the player can experience parenting a child with cancer from playing the indie game *That Dragon, Cancer* and understand adolescent identity, rebelliousness, and female friendships from playing *Life is Strange*. They can design the game to allow the player to control the PC to gradually interact with his companions, which can be a child, a sibling, a friend, or even a pet (see Consalvo & Begy, 2015) over the course of the game. Examples of these interactions have been discussed in Section 3.3.3 when participants 9–11 could connect to the characters when they can control Joel to constantly interact with Ellie throughout the game's narrative in *TLOU* and when participants 6–8 can control Lee Everett to constantly interact with Clementine in *TWD*. The constant interactions afforded to the players can allow them to empathize with the characters better as they explore the story and project their personalized narratives of the game. Beyond designing for paired relationships, game designers can consider integrating three-way relationships or more. For instance, the inclusion of a third character such as a pet can improve the relationship between the parent/caretaker and a child when they can learn how to take care of the pet together. In Sections 3.2.3 and 3.3.6, we have also discussed how participants 6–8's decision-making was influenced by Clementine who served as a moral anchor for the PC. Therefore, game designers can consider integrating a child character to nudge players on how to act in prosocial ways when they control their PC to interact with other characters in a game's narrative.

Table 4 summarizes the seven design principles that have been proposed for narrative games from the analysis of the player experience.

5. Limitations and future research

The study has some limitations that future research can address. First, the study's setting was not controlled. Some participants were allowed to play in the laboratory, while

Table 4. Design principles.

Design principles	Definition	Examples
Player-avatar dynamic	A PC that allows the player to both role-play and project their values, beliefs, morality, assumptions, and biases (Gee, 2007) onto it when playing the game.	In <i>TLOU</i> 's prologue, participant 9 played as Sarah followed by Joel. He projected the important value of family and the need to protect and save the PC's daughter when he played as Joel.
A strong interactive space for the player to explore	A strong interactive space can present ethical dilemmas for the player to project their values when playing the game. It can consist of rich visuals and narrative artefacts in the game environment that can bring out the believability aspect of the game's narrative. A strong interactive space can integrate elements of puzzle solving and mystery to arouse different emotions in the player to immerse them in the game's narrative.	Participant 2 explored <i>TW2</i>'s game environment in the prologue and chose to stop a commander who was about to set fire to the civilian's homes. Participant 6 explored <i>Bioshock</i>'s game environment, paused to reflect on it, and projected her personalized narrative of how Andrew Ryan's government was oppressing the poor people who were living in the slums in Apollo Square. Participants 6–8 explored the Saint John's dairy farm in <i>TWD Season 1 Episode 2</i> to uncover what was truly happening in the interactive space.
Placeholder for projecting an emergent narrative	A placeholder (player) character allows the player to project an emergent narrative or their personality, emotions, and values onto it.	In <i>BTS</i>, participant 1 projected Aiden as Jodie's protector and participant 2 projected Aiden as a mischievous entity. In <i>Bioshock</i>, participant 3 projected morally bad attributes to the PC when he hacked the turrets, flying drones, and the locked doors/safes.
Inclusive game design to enhance accessibility	Secondary PCs (or objects) use clearer multimodal cues to signal to the player when they can be used to perform specific actions to trigger narrative information (or be interacted with). Crucial narrative information should be given to the player when there is minimal gameplay.	In <i>TLOU</i>, clear multimodal indicators were used to let participants 9–11 know when the secondary PCs, NPCs, or objects can be interacted with to trigger narrative information. Participants 9–11 can focus on the narrative in <i>TLOU</i>'s prologue when they can only control Sarah to look left or right in Joel's car.
Interactive feedback loop to adapt to players' playstyle	The interactive story changes according to the players' degree of interactivity or passivity when playing the game.	<i>BTS</i> adapted to participants 1 and 2's level of involvement when playing the game.
QTEs and emotional connection to the characters	QTEs can allow players to form an emotional connection to the characters when they are integrated into the story, such as when there is a major development in the plot or when the PC suffered a major "loss."	In <i>TLOU</i> , participant 9 felt that the QTE involved him in the story and allowed him to feel the emotional connection to Ellie when his PC, Joel suffered a major injury and became dependent on Ellie.
More complex options to romance/relationships in narrative games	The romance options in narrative game design can be made more in-depth by allowing more time for the PC to develop a relationship with the NPC. The PC's relationship with the NPC can change from the interaction with multiple NPCs. Relationship dynamics can be broadened beyond romance to include parent/caretaker-child relationships, friendships, sibling relationships, LGBT relationships, and child-pet relationships.	The PC can work together with the NPC in a series of loyalty missions like those in <i>ME</i> to strengthen the relationship with the NPC. The improvement of a romantic relationship with an NPC can worsen a romantic relationship with another NPC. Participants 6–8 demonstrated how they behaved in prosocial ways in front of Clementine when they made decisions in the context of the game. For instance, they chose to save Lily's father in the meat locker in <i>TWD Season 1 Episode 2</i>.

others were allowed to play and record their selected game(s) at home. This introduces a host of potential variables in the setting and context of play that could bias the study's results. For instance, participant 6 mentioned through the email correspondence that if she were to play *TLOU* under the research environment in the laboratory, she did not think that she will take her time to scavenge for raw materials and craft items and she would want to get on with the game. Similarly, participant 11 mentioned that he would take his time to explore the game world in *TLOU* more thoroughly if he were to play at home. Future research should control the study's setting such that all participants were either instructed to play at home or in the laboratory. Second, there is a male gender bias in the study due to the snowball and convenience sampling method used to recruit participants. Future research should aim to recruit participants with a balanced gender ratio to address the gender bias. Third, there was no attempt to control the genre differences in the games used in the study. Although all the games are narrative-driven, the games are from different genres with different control configurations, camera angles,

gameplay mechanics, and so on. A future study might narrow the number of games presented to participants such that different groups of participants interact with one game to control the variables present in the study to make the findings clearer and more generalizable.

Fourth, the study did not control for players' narrative preferences, either in participant selection or in data coding and we have recruited participants who liked narrative games and participants who preferred to play games for the gameplay before the study. A future study can control the players' narrative preferences such that either only participants who like narrative games or participants who prefer gameplay games are recruited. Fifth, even though the small sample size of 12 participants in this study provides us with detailed descriptions of the participants' interactional patterns in their games played for the study, the findings may not be representative. A future study can recruit a larger sample size to ensure that the findings are more representative.

Sixth, the use of the think-aloud protocol to elicit the participants' thought processes in the form of verbalizations has

some limitations. Players might not think-aloud as expected when they were not used to doing that when playing the game and instead focused on expressing their reactions to the game. Players were also silent when they focused on playing the game. When they did not think-aloud, the researcher would then have to review the recordings with them and check with them about their thought processes during the interviews using the retrospective think-aloud protocol. Seventh, this study did not test games without defined narratives such as sandboxes because it focuses on narrative games with defined narratives. A future study can compare games with defined narratives with games without defined narratives to understand how emergent narrative can be generated when the players interact with the other characters in single player games or with other players in multi-player games (Toh & Lim, 2021). Finally, a future study can also consider using a narrative framework as a guiding principle for game designers, especially one dealing with rich and immersive narrative gameplay, for them to use. The narrative framework may consist of the three-act structure like those found in traditional storytelling (e.g., Ip, 2011) or structuralist narrative theories, such as those from Valdimir Propp's pioneer work *Morphology of the Folktale* that can be used as a guide to generate narrative structures or narrative events in video games (e.g., Lindley, 2005). However, this might require interviews with narrative designers and game writers to understand how they adapt and integrate these narrative frameworks for use in narrative game design.

6. Conclusion

In this article, we have analyzed the player experience of narrative games using grounded theory by focusing on an inductive study of the participants' data, including their interviews and video recordings with their think-aloud verbalizations and reflections, to derive the four interactional patterns. Based on the analysis, we have suggested design implications that game designers can adopt in future designs of games for interactive storytelling. The proposed human computer interaction dimensions in the form of the four interactional patterns can be used by game designers as a basis to evaluate narrative game design and to generate narrative game design principles to improve the player experience. Game designers and user experience researchers can also build and refine the four interactional patterns as they examine more narrative games.

Researchers and game designers can also use the proposed interactional patterns in user experience studies to analyze how the players' emotions change as their interactional patterns in narrative games change in the different parts of the game. They can correlate the players' emotions with the interactional patterns to understand how their emotions fluctuate when playing the game. Researchers and game designers can examine the quantity and types of emotions players are experiencing in the four interactional patterns using emotion recognition technology (e.g., Ioannou et al., 2005). More objective instruments such as the galvanic skin conductance can also be used to track the players'

engagement during narrative gameplay to examine which parts of the game players are engaged with and which parts of the game players are disengaged with. The empirical data collected can be used to evaluate narrative game design to improve aspects of the game to immerse the player in the game, allow them to form emotional connections to the characters, and to enhance their understanding and appreciation of the game's narrative.

Notes

1. In *BioShock* Remastered Directors Commentary Episode 5 (0:50-1:26: <https://tinyurl.com/Bioshockmodels>), Ken Levine and Shawn Robertson mentioned that there were time and budget constraints that contributed to the reuse of character models in *Bioshock*.
2. <https://tinyurl.com/mocapTW2>
3. However, participant 2 mentioned in the recordings that the animations are reused in the cutscenes that show the sexual encounters between the PC and the prostitutes at the Kaedweni brothel in chapter 2 of *TW2* as he explored the different options (see endnote 7). Participant 12 mentioned in the recordings that the graphics in the cutscene that show the sexual encounter between the PC and the NPC, Cynthia in chapter 3 of *TW2* is not very good (see endnote 10).
4. <https://tinyurl.com/P6ApolloSquare>
5. <https://tinyurl.com/P2niceguy>
6. <https://tinyurl.com/P12succubus>
7. <https://tinyurl.com/P2brothel>
8. <https://tinyurl.com/P12letsSiladie>
9. <https://tinyurl.com/P12killsdragon>
10. <https://tinyurl.com/Cynthiaforced>
11. <https://tinyurl.com/Joelimpaled>
12. <https://tinyurl.com/P12acceptstriss>
13. <https://tinyurl.com/bioshockghosts>
14. <https://tinyurl.com/P6Jasmine>
15. <https://tinyurl.com/P9plotdevice>
16. <https://tinyurl.com/Henseltbetrayal>
17. <https://tinyurl.com/P2projectsrape>
18. <https://tinyurl.com/Henselttrapesves>
19. <https://tinyurl.com/Rochekillshenselt>
20. <https://tinyurl.com/P2Henselt>
21. <https://tinyurl.com/P2TW2QTE>
22. <https://tinyurl.com/P12Dethmold>
23. Participant 7's think-aloud: <https://tinyurl.com/P7crawfordthinkaloud>
24. 15:40 and 18:00 in the link: <https://tinyurl.com/TWDCrawford>
25. A full list of game accessibility guidelines can be accessed at <https://gameaccessibilityguidelines.com/full-list/>

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